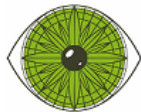
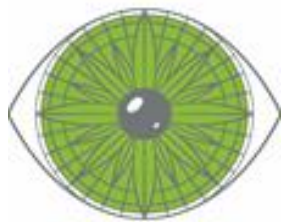


# Graphic display of (geospatial) data

**Jacques Gautier**

IGN-ENSG, LaSTIG, Equipe GeoVIS, DCAIG

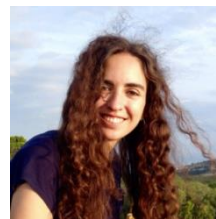
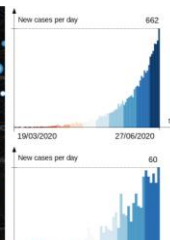
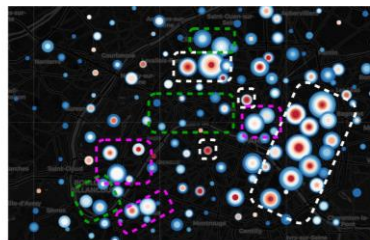
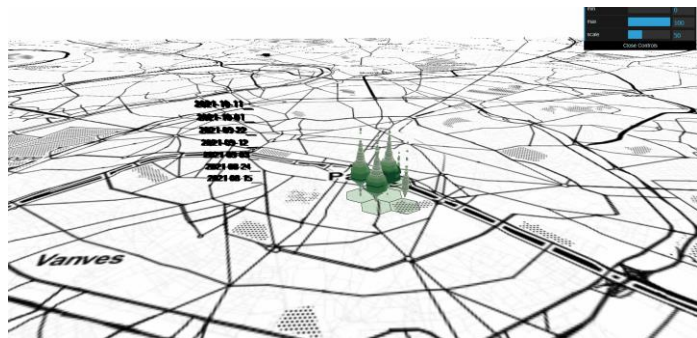
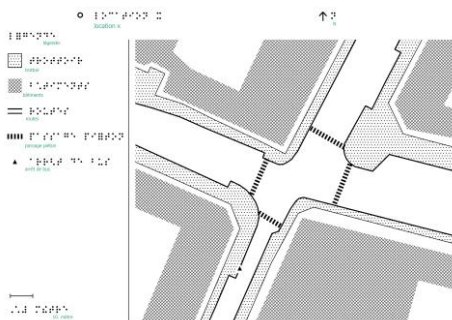
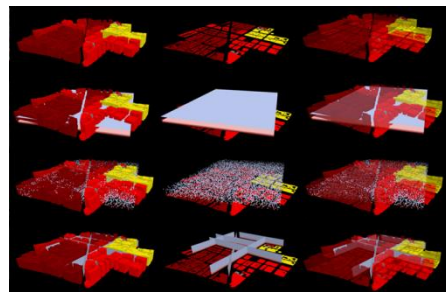




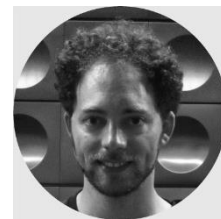
# GEOVIS Team : visualization, interaction, immersion



➤ Provide knowledge, methods, and geovisualization tools for the analysis and communication of spatiotemporal phenomena.



María Jesús  
LOBO



Jacques  
GAUTIER

# Visualization

Visualization refers to a set of methods, computer-based, to help users perform different types of tasks more efficiently, using visual representations of data.

*Adapted from Munzner, 2014*

# Visualization

Visualization refers to a set of methods, computer-based, to help users perform different types of tasks more efficiently, using **visual representations** of data.

*Adapted from Munzner, 2014*

# Why using a visual representation ?

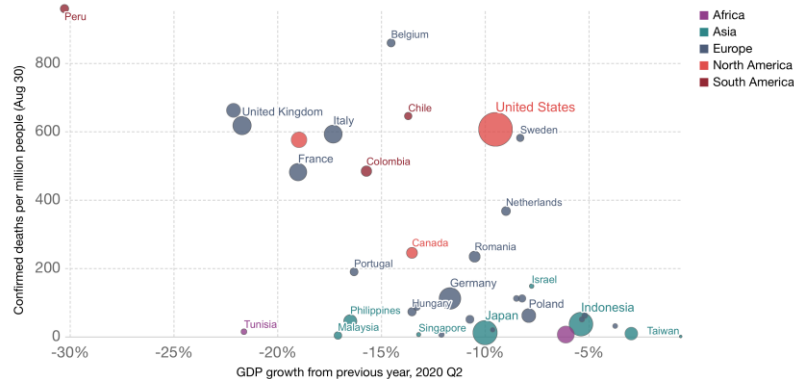
| Country        | Total confirmed deaths due to COVID-19 per million people deaths per million | GDP growth from previous year, 2020 Q2 percent |
|----------------|------------------------------------------------------------------------------|------------------------------------------------|
| Peru           | 867.62                                                                       | -30.2%                                         |
| Spain          | 620.49                                                                       | -22.1%                                         |
| United Kingdom | 611.29                                                                       | -21.7%                                         |
| Tunisia        | 6.43                                                                         | -21.6%                                         |
| France         | 468.83                                                                       | -19.0%                                         |
| Mexico         | 494.98                                                                       | -19.0%                                         |
| Italy          | 586.70                                                                       | -17.3%                                         |
| Malaysia       | 3.86                                                                         | -17.1%                                         |
| Philippines    | 31.20                                                                        | -16.5%                                         |
| Portugal       | 178.29                                                                       | -16.3%                                         |
| Colombia       | 374.66                                                                       | -15.7%                                         |
| Belgium        | 853.44                                                                       | -14.5%                                         |
| Chile          | 584.90                                                                       | -13.7%                                         |
| Hungary        | 63.56                                                                        | -13.5%                                         |
| Canada         | 241.45                                                                       | -13.5%                                         |
| Austria        | 81.39                                                                        | -13.3%                                         |
| Singapore      | 4.62                                                                         | -13.2%                                         |
| G7             |                                                                              | -12.1%                                         |
| Slovakia       | 6.04                                                                         | -12.1%                                         |
| Cyprus         | 23.98                                                                        | -11.9%                                         |
| Germany        | 110.94                                                                       | -11.7%                                         |
| Czech Republic | 39.31                                                                        | -10.7%                                         |
| Romania        | 183.96                                                                       | -10.5%                                         |
| Japan          | 9.99                                                                         | -10.0%                                         |
| Latvia         | 18.03                                                                        | -9.6%                                          |
| United States  | 552.20                                                                       | -9.5%                                          |
| Netherlands    | 362.71                                                                       | -9.0%                                          |
| Denmark        | 107.73                                                                       | -8.5%                                          |
| Sweden         | 577.86                                                                       | -8.3%                                          |
| Bulgaria       | 87.07                                                                        | -8.2%                                          |
| Poland         | 53.69                                                                        | -7.9%                                          |
| Israel         | 104.67                                                                       | -7.8%                                          |
| Nigeria        | 4.90                                                                         | -6.1%                                          |
| Indonesia      | 26.55                                                                        | -5.4%                                          |
| Norway         | 48.70                                                                        | -5.3%                                          |
| Finland        | 60.46                                                                        | -5.2%                                          |
| Lithuania      | 31.59                                                                        | -3.7%                                          |
| South Korea    | 6.30                                                                         | -3.0%                                          |
| Taiwan         | 0.29                                                                         | -0.6%                                          |
| China          | 3.28                                                                         | 3.2%                                           |

« External representation: replace cognition with perception »  
Munzner, 2014

## Economic decline in the second quarter of 2020 vs rate of confirmed deaths due to COVID-19

The vertical axis shows the number of COVID-19 deaths per million, as of August 30. The horizontal axis shows the percentage decline of GDP relative to the same quarter in 2019. It is adjusted for inflation.

Our World in Data



Source: European CDC, Eurostat, OECD and individual national statistics agencies

CC BY

Note: Limited testing and challenges in the attribution of the cause of death means that the number of confirmed deaths may not be an accurate count of the true number of deaths from COVID-19. Data for China is not shown given the earlier timing of its economic downturn. The country saw positive growth of 3.2% in Q2 preceded by a fall of 6.8% in Q1.

Allows for the direct reading of information

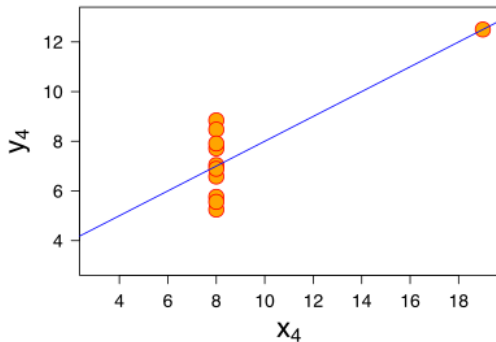
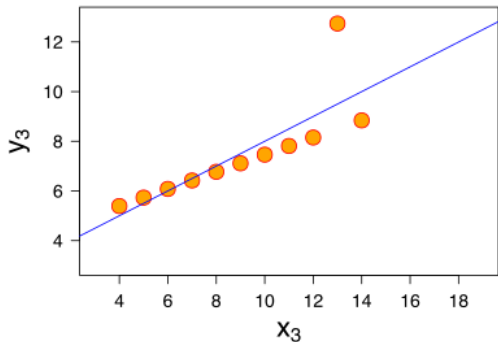
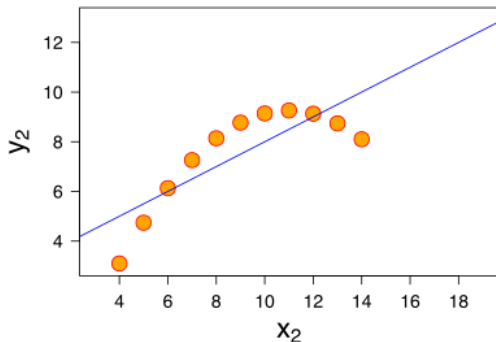
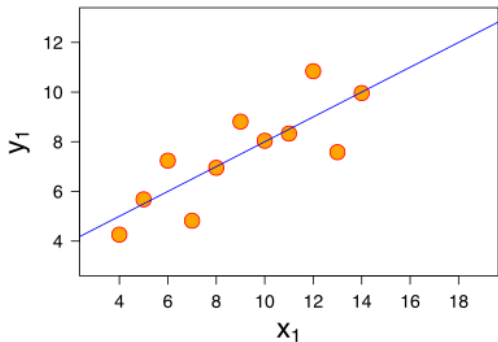
There is no sequential decoding of different groups of symbols

# **Why using a visual representation ?**

... instead of calculating (statistical) indicators ?

# Why using a visual representation ?

... instead of calculating (statistical) indicators ?



## Anscombe Quartet

Same basic stat indicators:

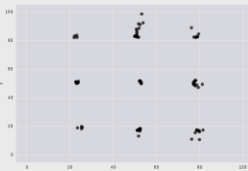
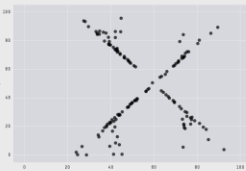
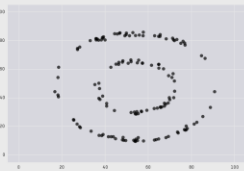
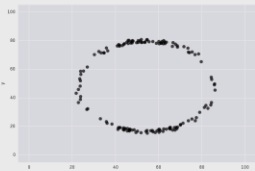
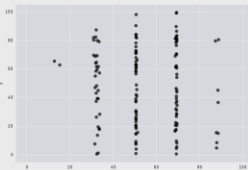
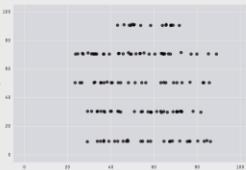
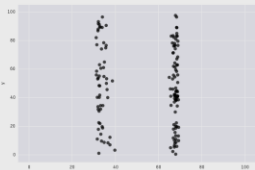
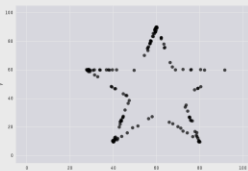
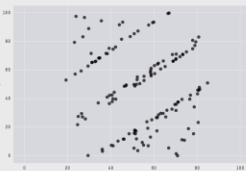
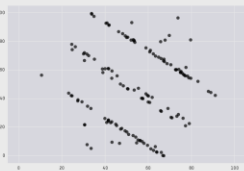
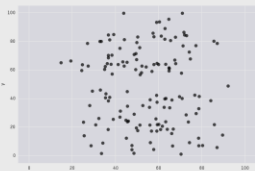
- mean
- variance
- correlation
- regression line

# Why using a visual representation ?

... instead of calculating (statistical) indicators ?



X Mean: 54.26  
Y Mean: 47.83  
X SD : 16.76  
Y SD : 26.93  
Corr. : -0.06



Same basic stat indicators:

- mean
- variance
- correlation
- regression line

AUTODESK Research

<https://www.autodeskresearch.com/publications/samestats>.

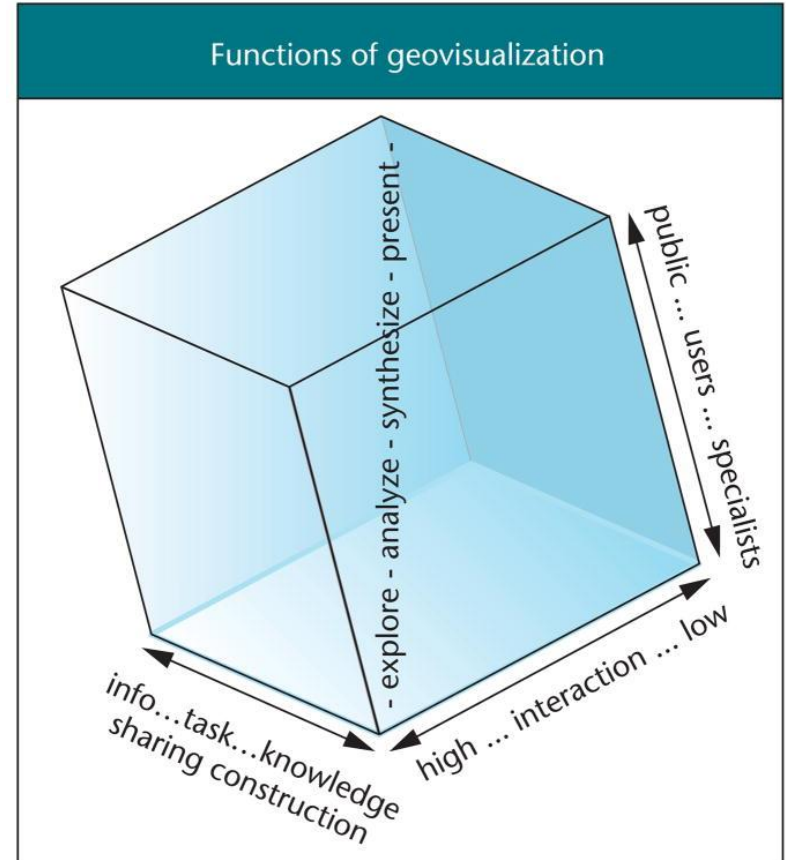


# Why using a visual representation ?

Visualizing directly data allows:

- Identifying unsuspected patterns in the data
- Better interpret suspected patterns in the data

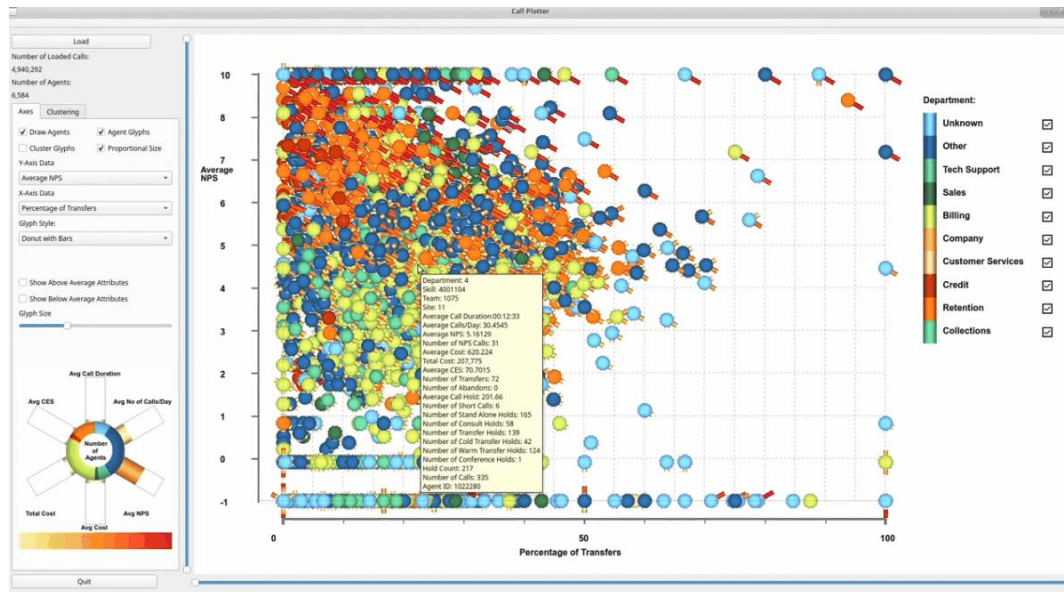
# The objectives of visualizations



# Visualization-related domains

## Dataviz

### Data Visualization



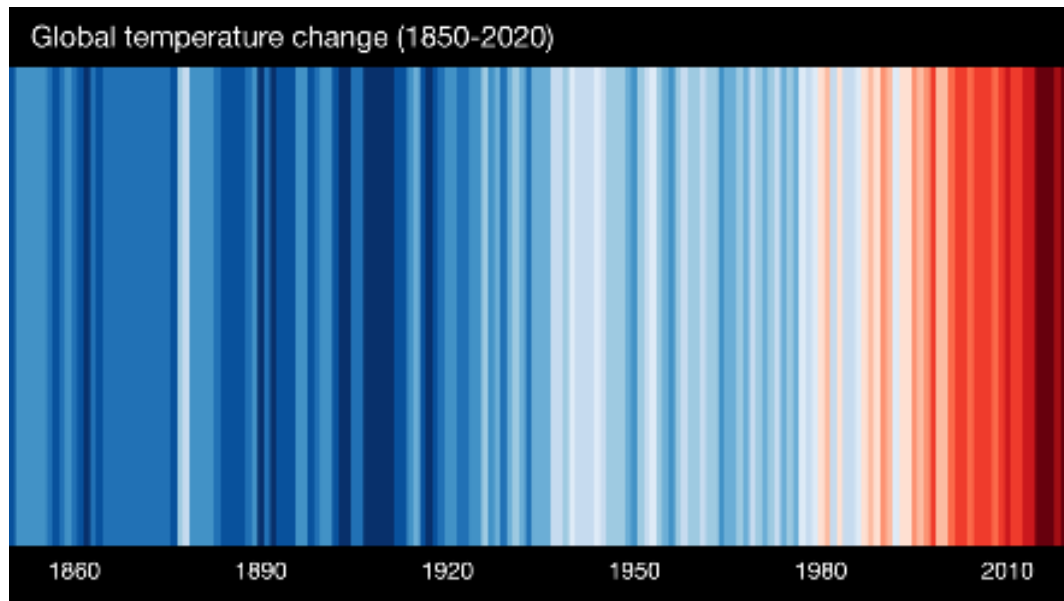
# Visualization-related domains

## **Dataviz**

*Data Visualization*

## **Infoviz**

*Information Visualization*



# Visualization-related domains

## **Dataviz**

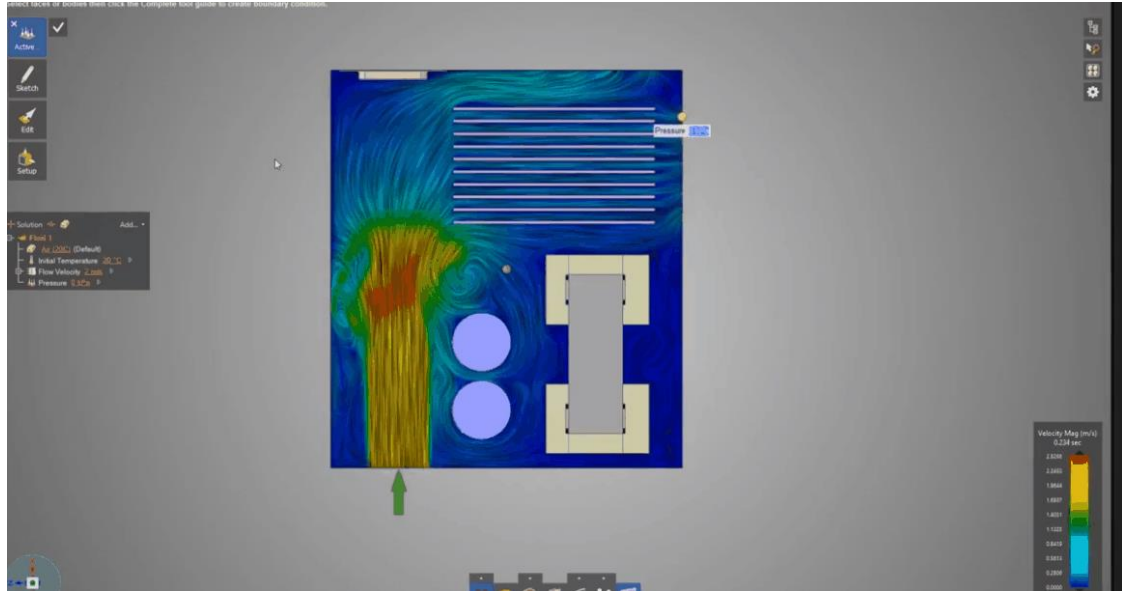
*Data Visualization*

## **Infoviz**

*Information Visualization*

## **Sciviz**

*Scientific Visualization*



# Visualization-related domains

## Dataviz

*Data Visualization*

## Infoviz

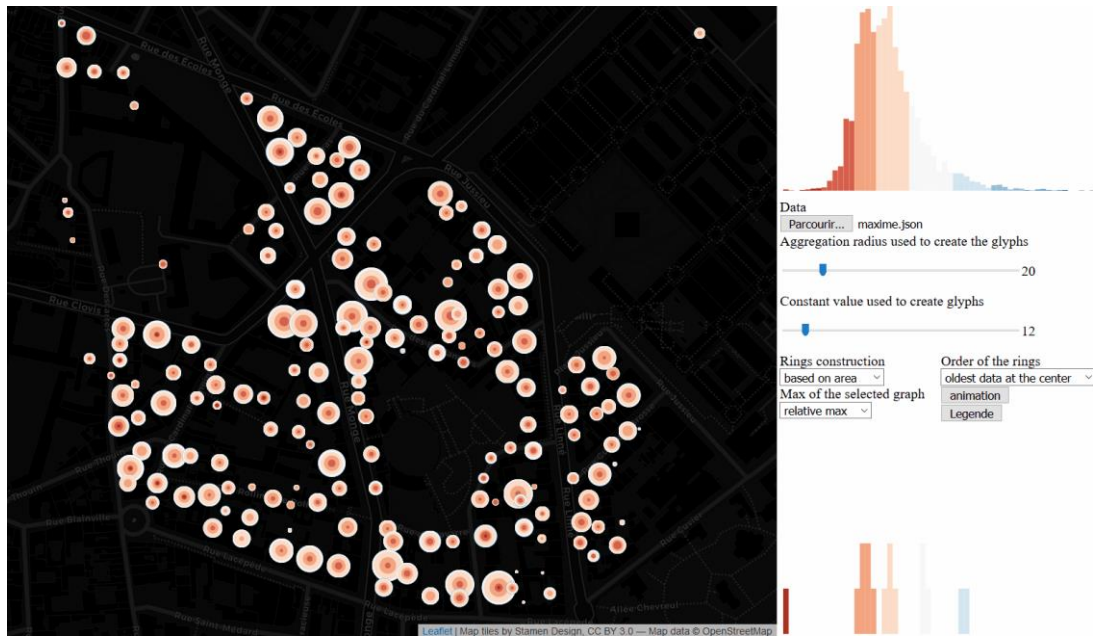
*Information Visualization*

## Sciviz

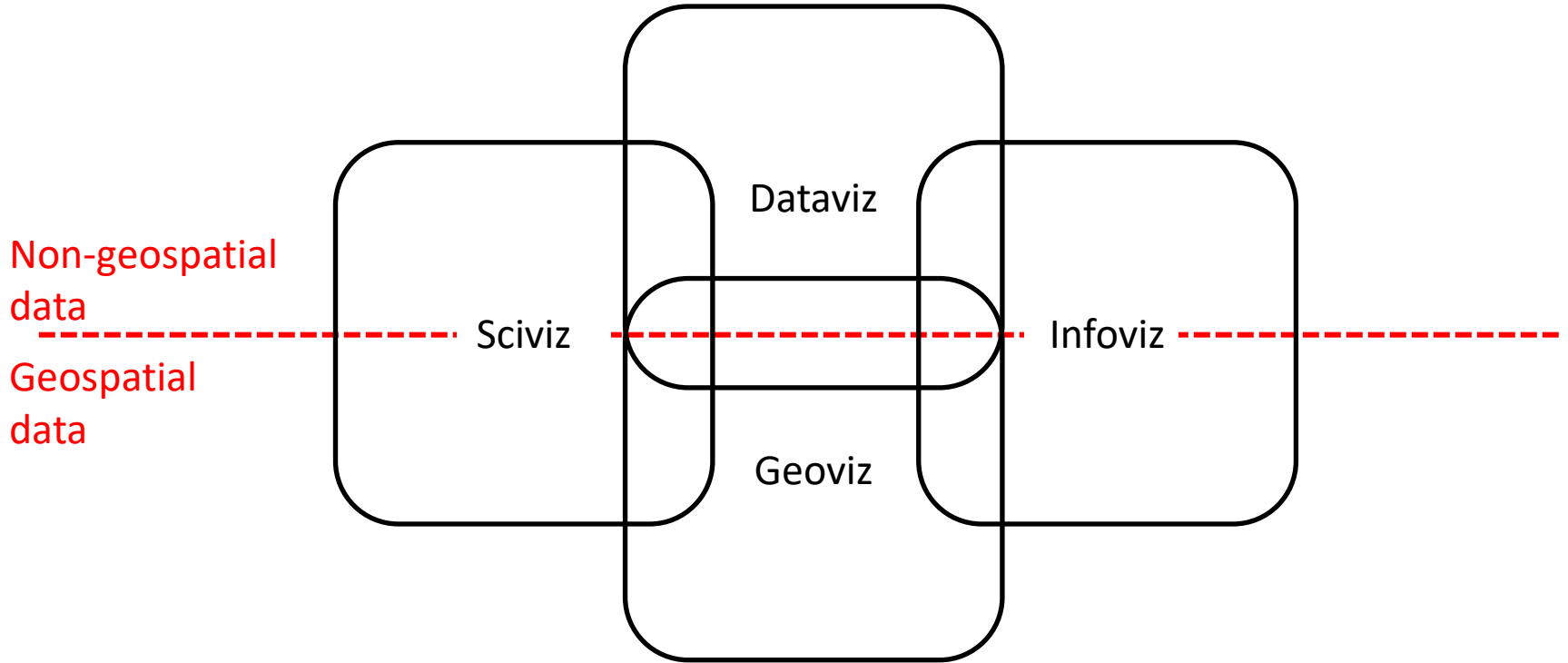
*Scientific Visualization*

## Geovisualization

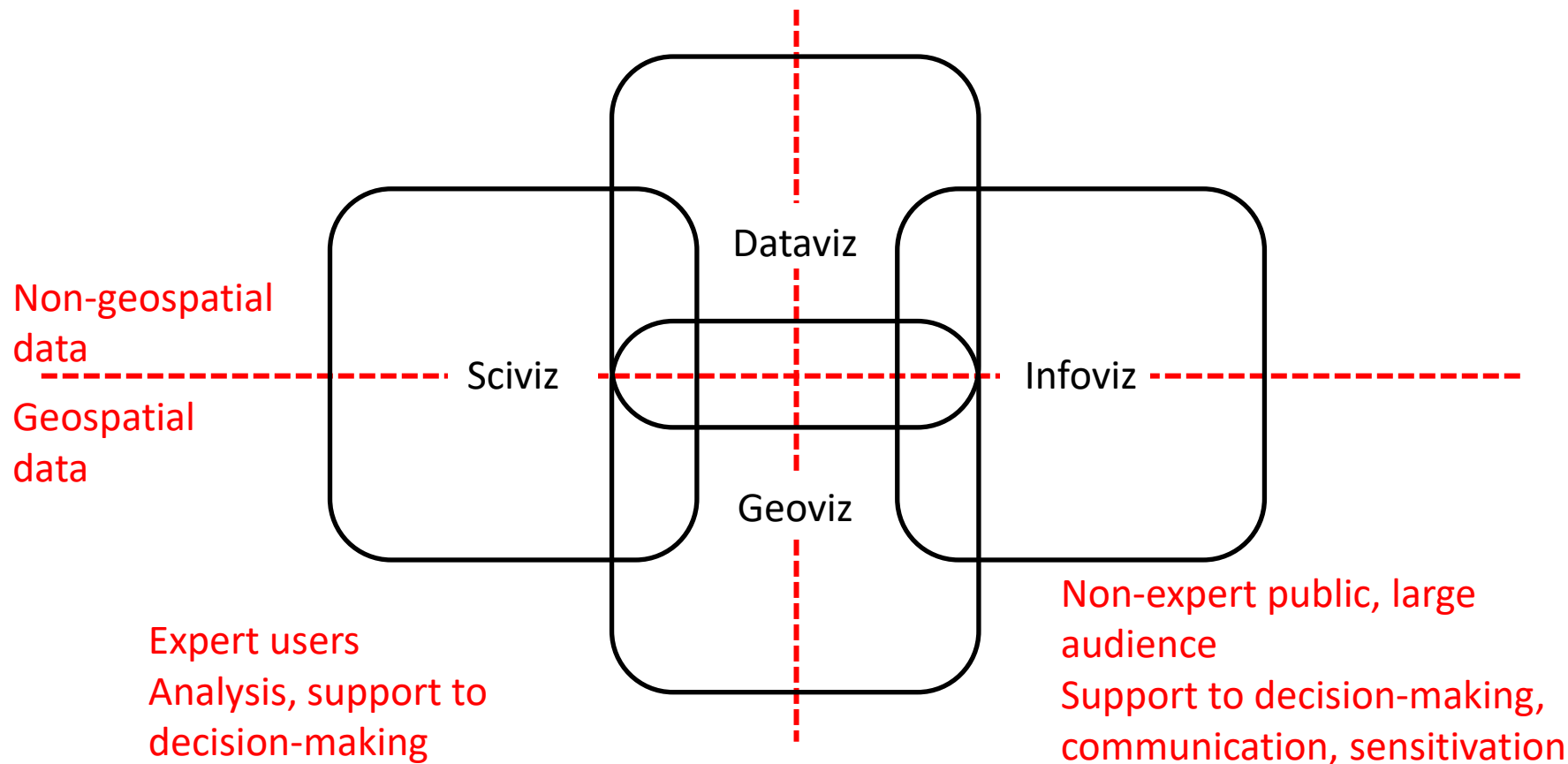
*Geospatial data visualization*



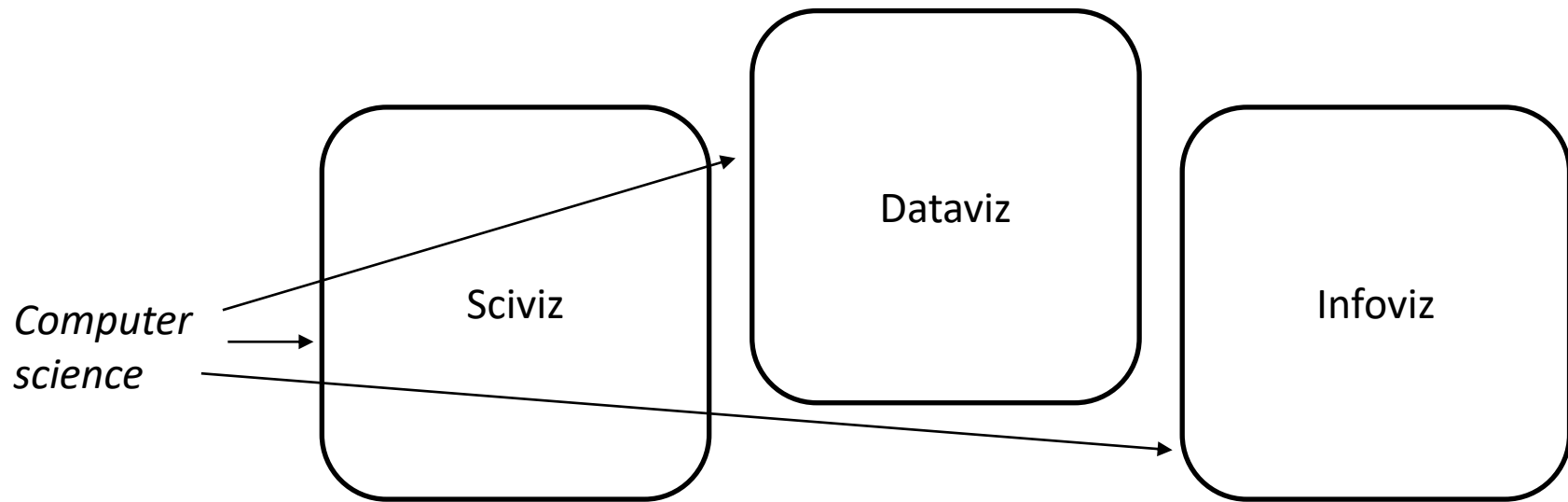
# Visualization-related domains

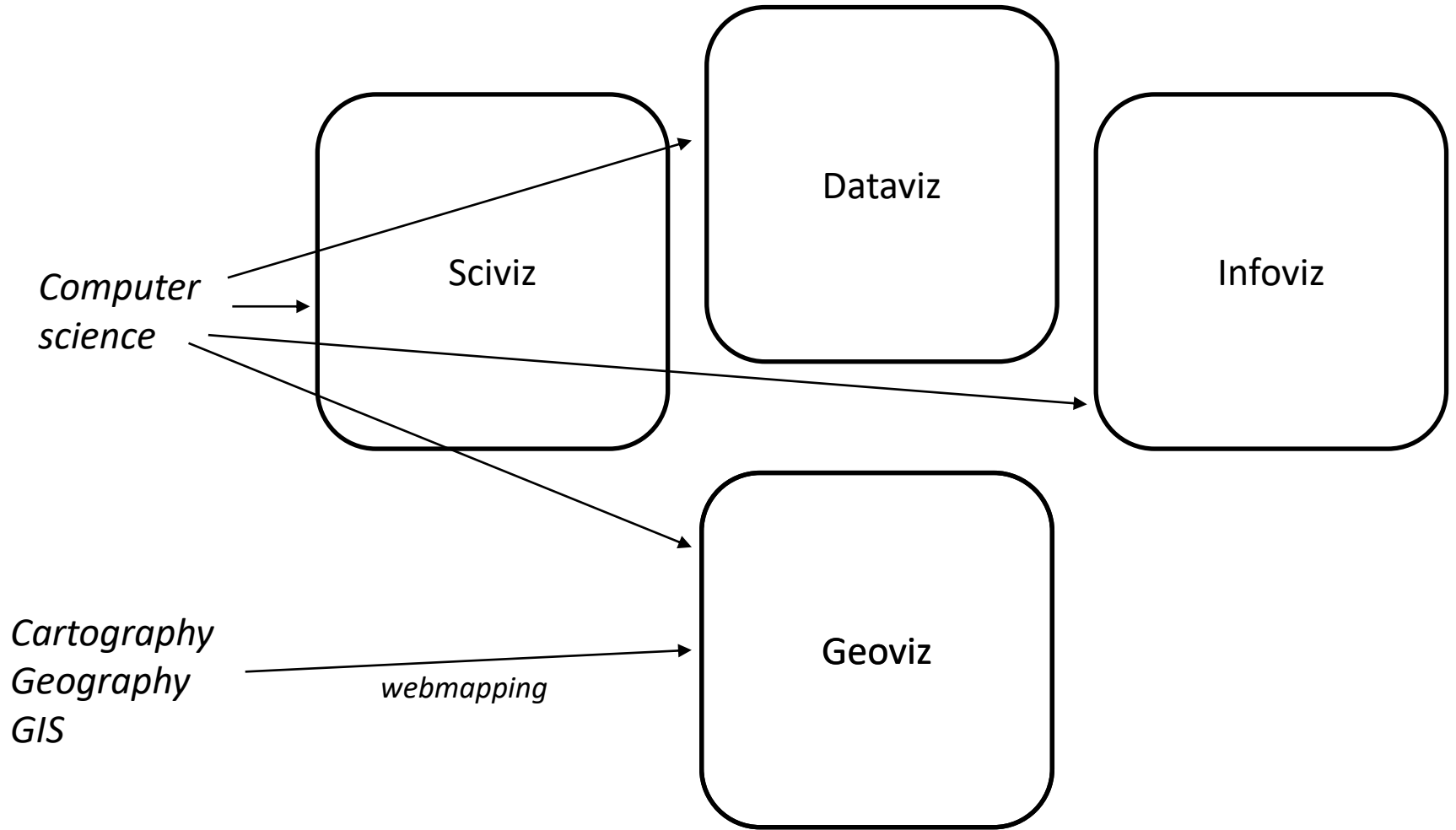


# Visualization-related domains









*Computer  
science*

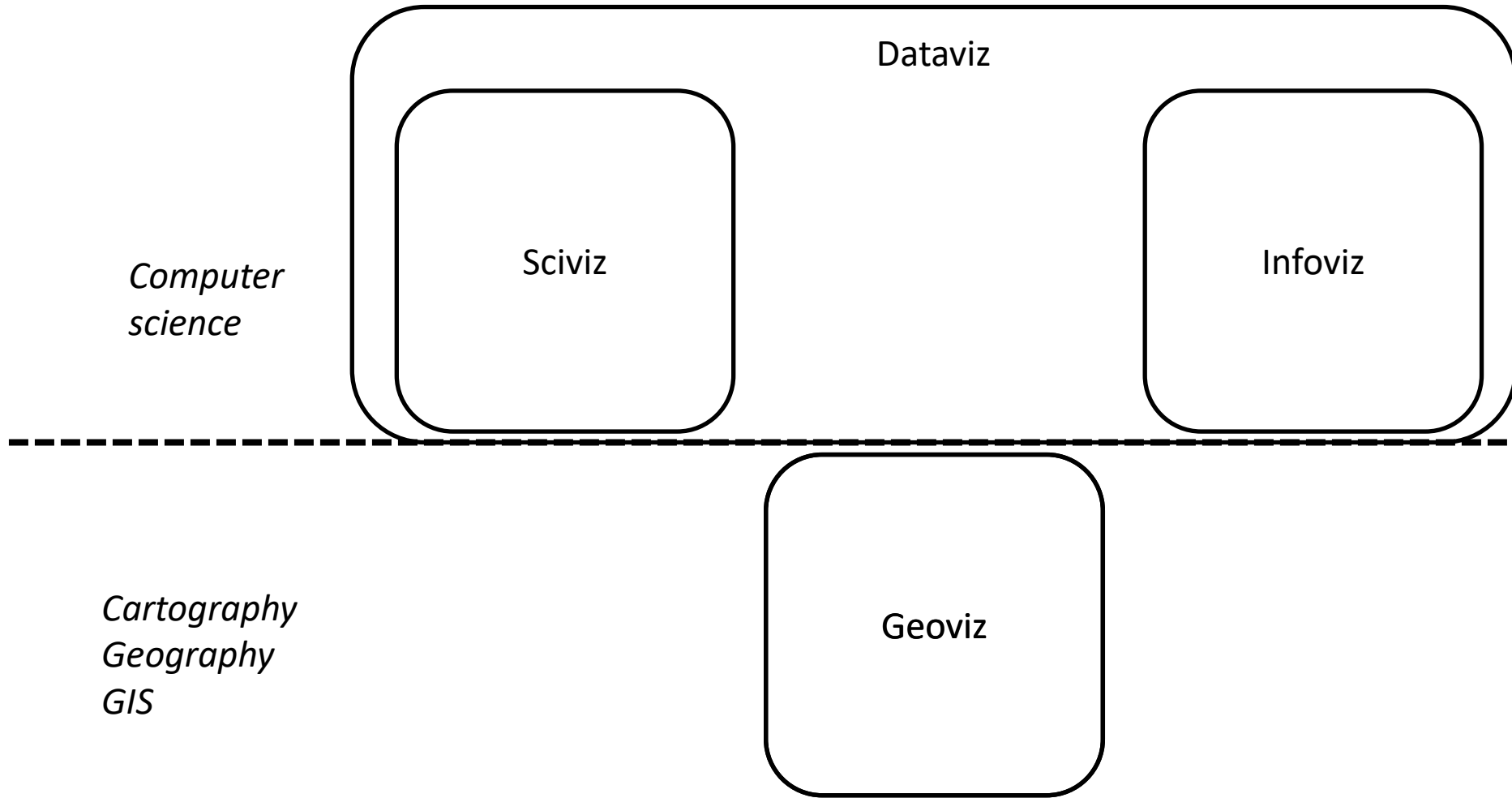
Sciviz

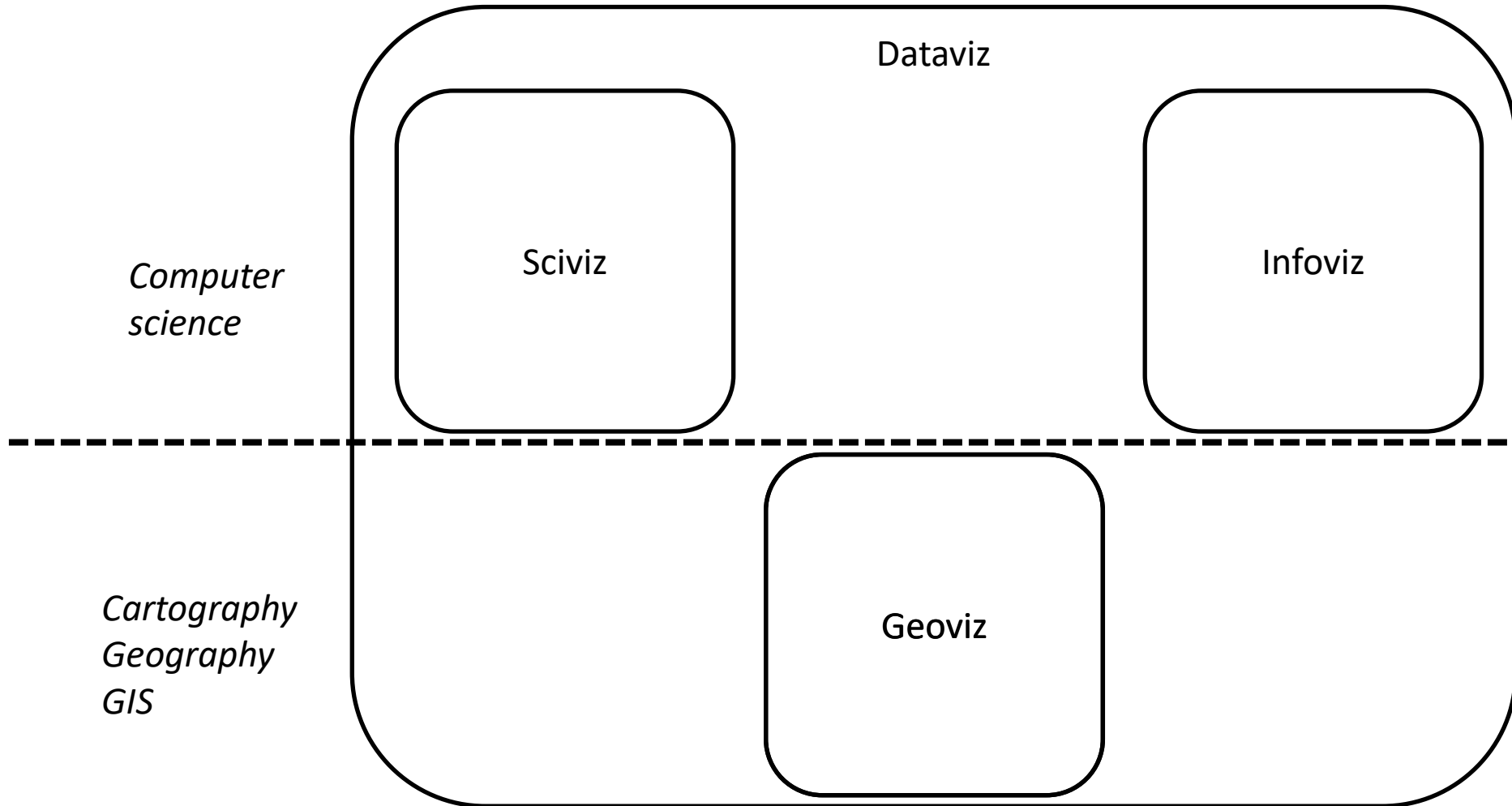
Dataviz

Infoviz

*Cartography  
Geography  
GIS*

Geoviz







DOCS    HELP    LOGIN

Dataset

ncov

gisaid

global

Date Range

2019-12-18

2021-10-29

▶ PLAY

↺ RESET

Color By

Clade

Filter Data

Type filter query here...

Tree Options

Layout

 RECTANGULAR

 RADIAL

 UNROOTED

 CLOCK

 SCATTER

Branch Length

TIME

DIVERGENCE

☐

Show confidence intervals

Branch Labels

clade

Tip Labels

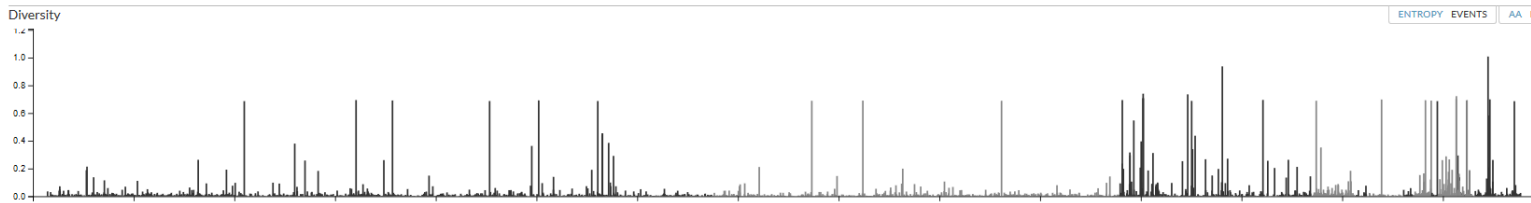
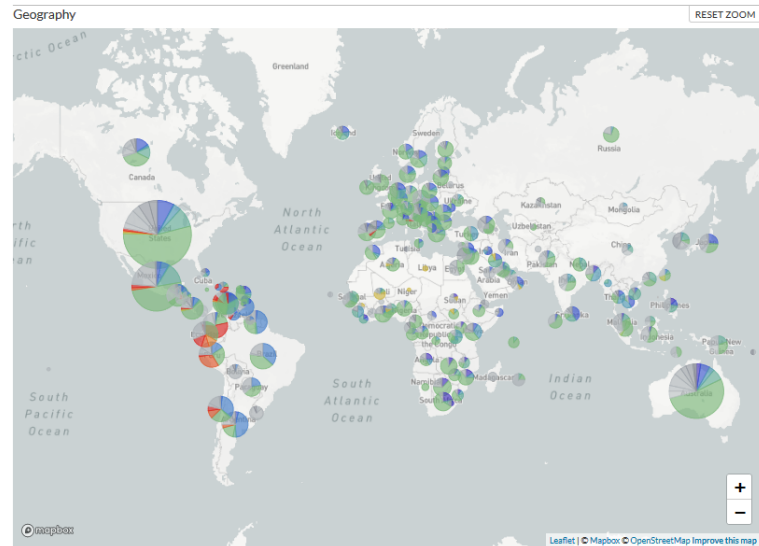
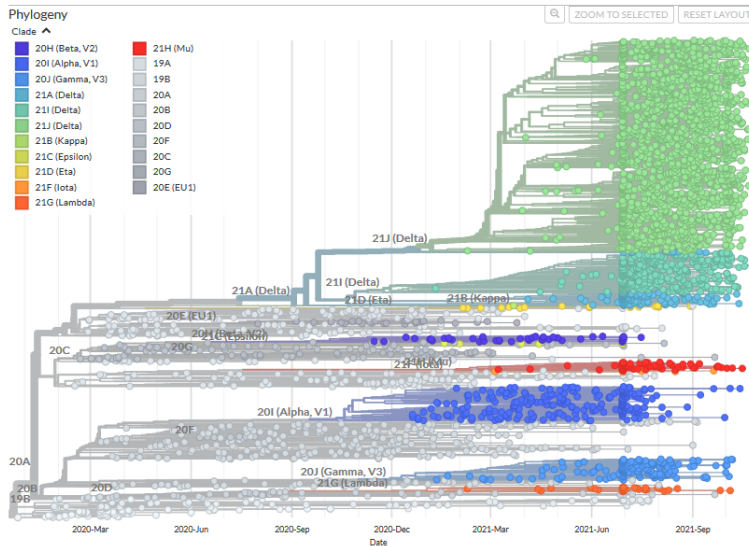
Sample Name

Second Tree

Select...

Map Options

Built with [nextstrain/ncov](#). Maintained by the [Nextstrain team](#). Enabled by data from [GISAID](#)



# Visualization-related domains

Several domains...

...with fuzzy borders...

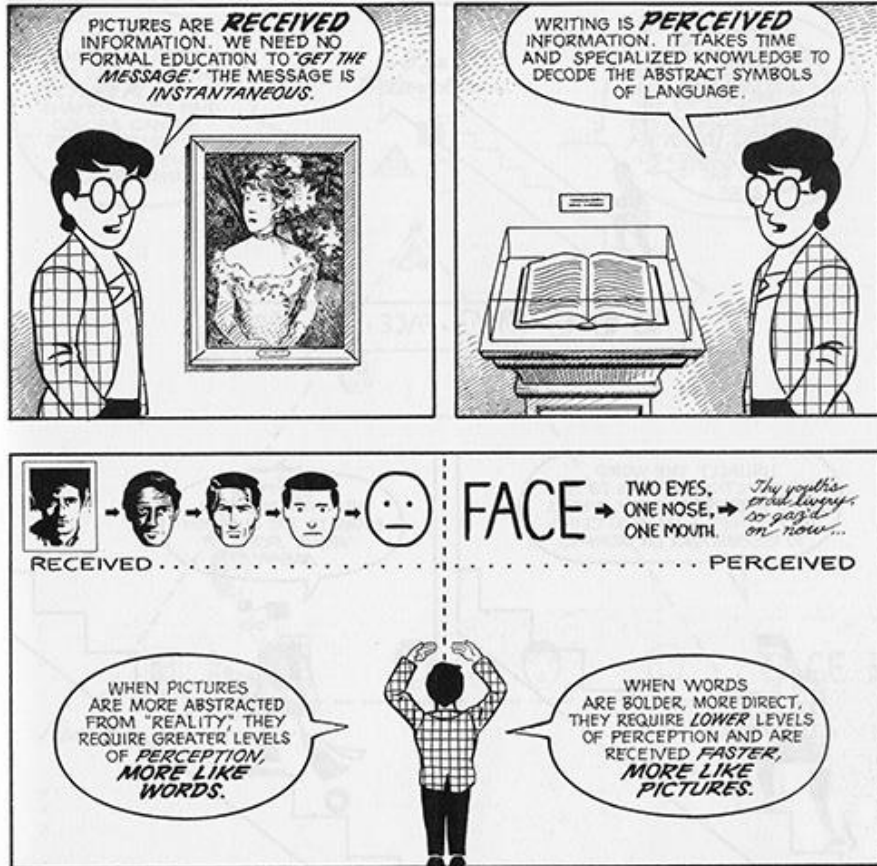
...which could be defined by use-cases, visualization approaches, end-users, but also by their communities of visualization producers...

...and with common problematics

Using a symbolic language



# Graphic abstraction and symbols



Scott McCloud, Understanding Comics.

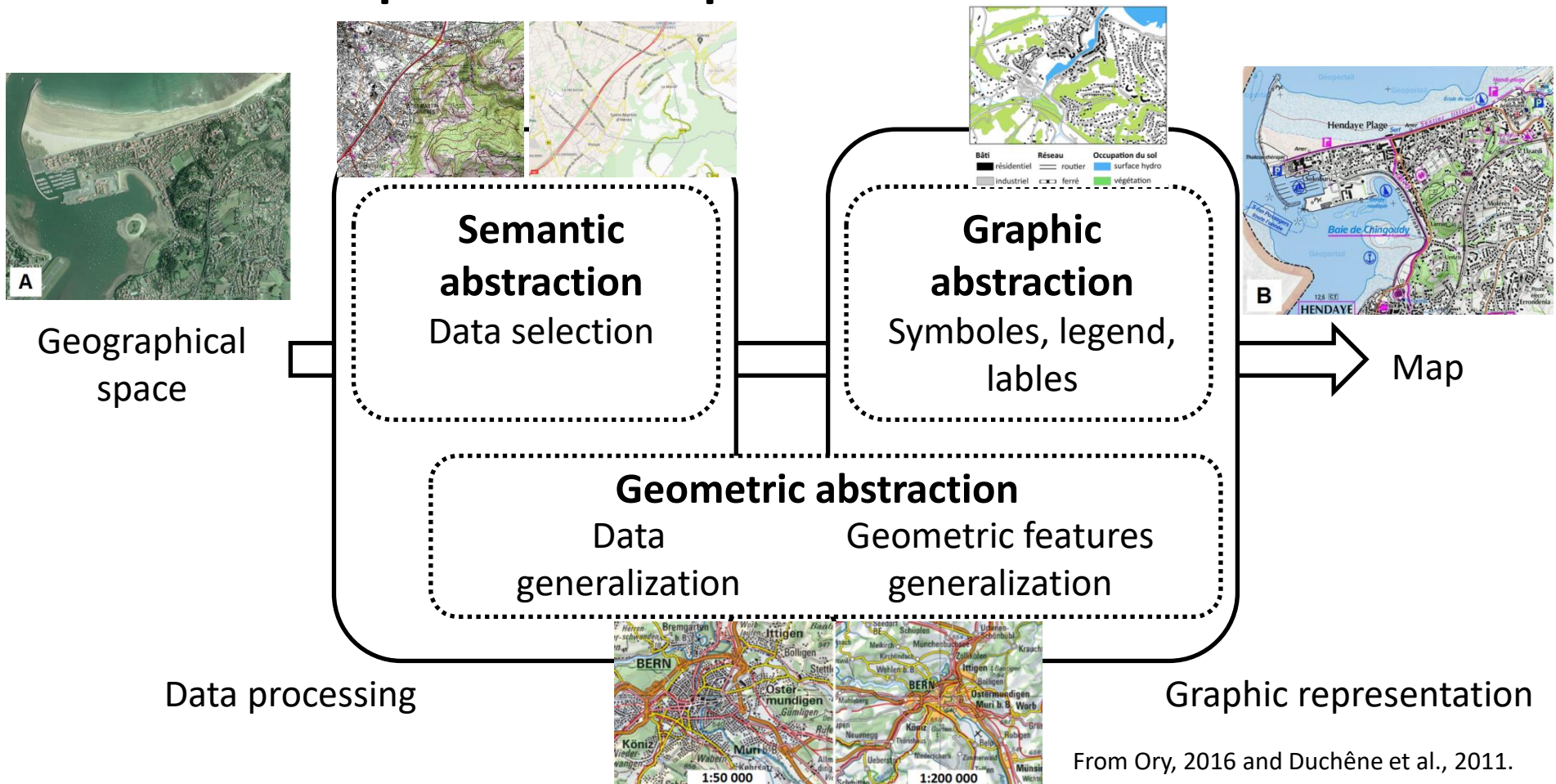
A visual representation is more effective for conveying information quickly.

But an image is polysemic, does not have a single meaning

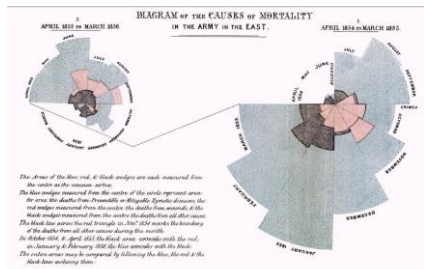
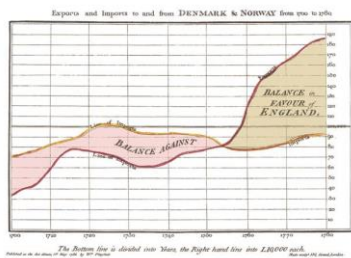
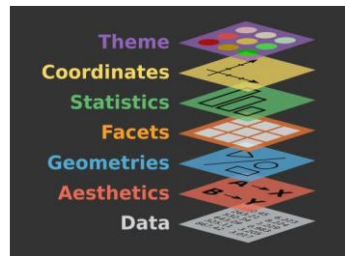
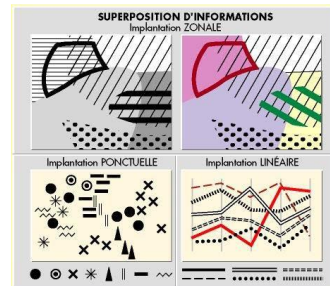
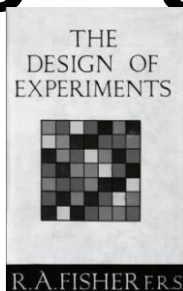
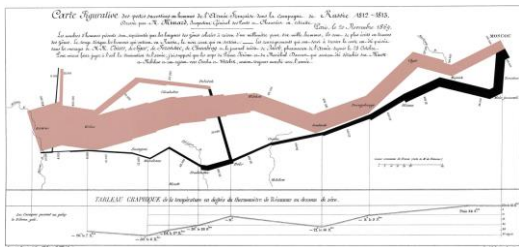
Then, how to transmit a specific information through an image ?

We use abstract representations, using a symbolic languages: a list of normalized symbols following rules dictated by visual perception.

# An example: a map

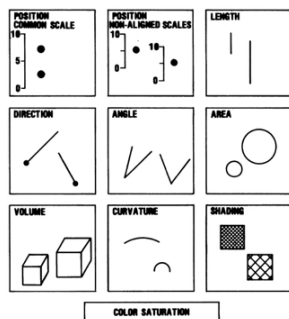
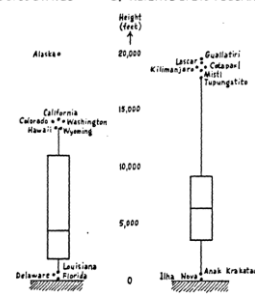


# Symbolic language



A) HEIGHTS OF 50 STATES

B) HEIGHTS OF 219 VOLCANOE



**William Playfair**

**Charles Minard**

**Florence Nightingale**

**Ronald Fisher**

**John Tukey**

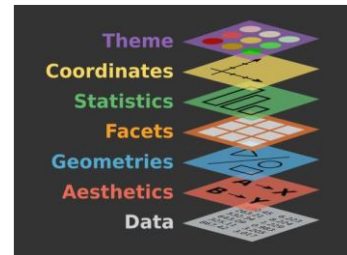
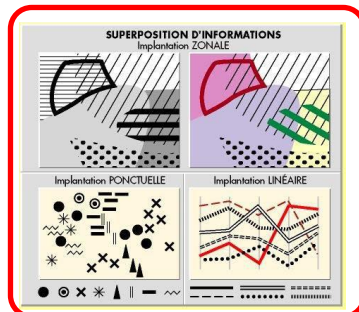
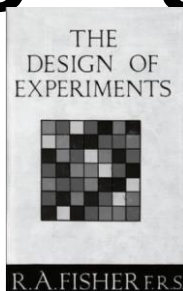
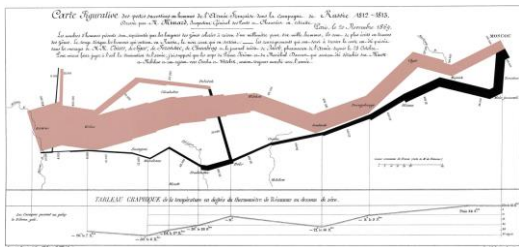
**Jacques Bertin**

**William Cleveland**

**Leland Wilkinson**

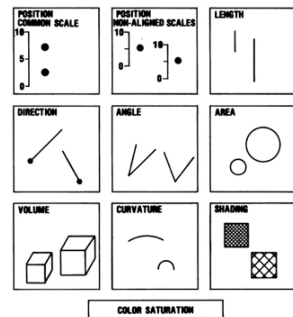
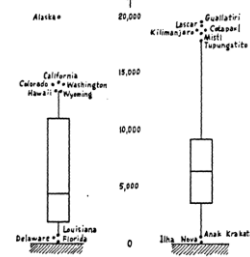
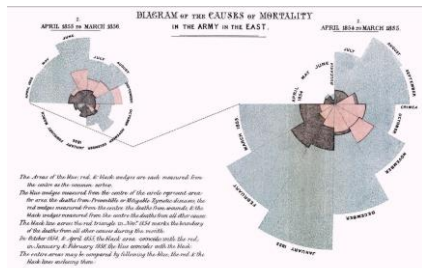
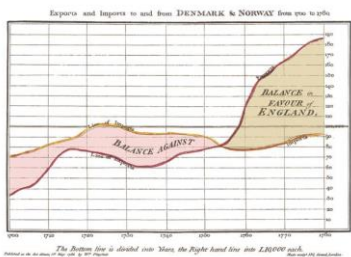
(1759-1823) (1781-1870) (1820-1910) (1890-1962) (1915-2000) (1918-2010) (1943- ) (1944-2021)

# Symbolic language



A) HEIGHTS OF 50 STATES

B) HEIGHTS OF 219 VOLCANOE



**William Playfair**

**Charles Minard**

**Florence Nightingale**

**Ronald Fisher**

**John Tukey**

**Jacques Bertin**

**William Cleveland**

**Leland Wilkinson**

(1759-1823)

(1781-1870)

(1820-1910)

(1890-1962)

(1915-2000)

(1918-2010)

(1943- )

(1944-2021)

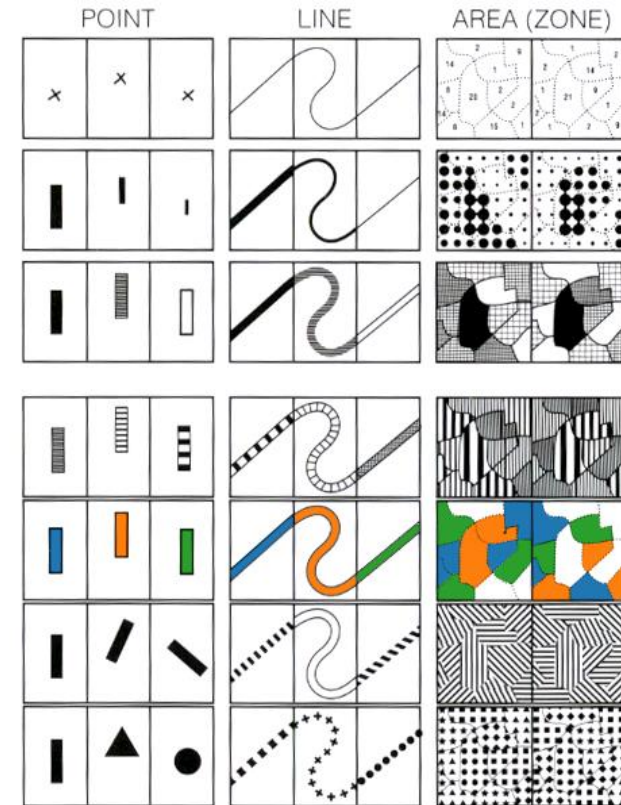
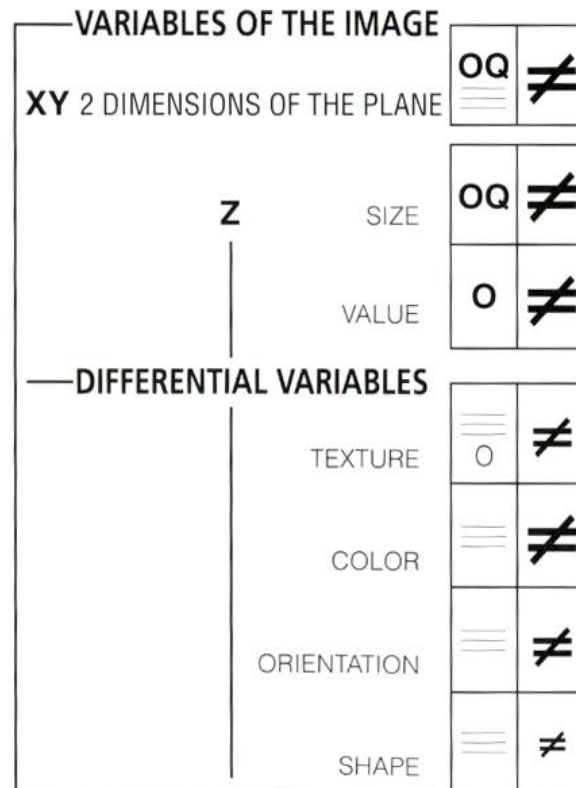
# Symbolic language

## Bertin 1967

# Graphic semiology

*Set of rules governing the construction of symbol systems or languages allowing the graphic transcription of information*

# Catherine Domingues, 2008





# Properties of visual variables

## Association

Capacity to identify a property common to different objects

## Differentiation (selection)

Capacity to identify an object (or a category of objects) among others

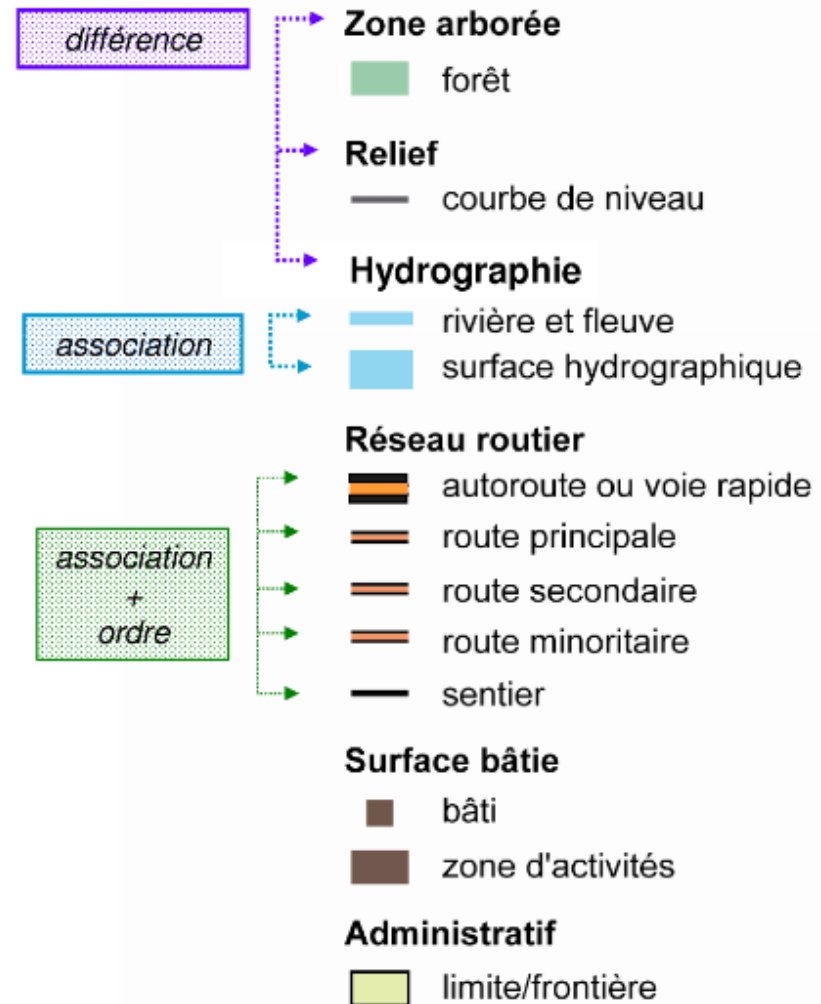
## Order

Capacity to identify a hierarchy among objects (or categories of objects)

## Quantity

Capacity to estimate

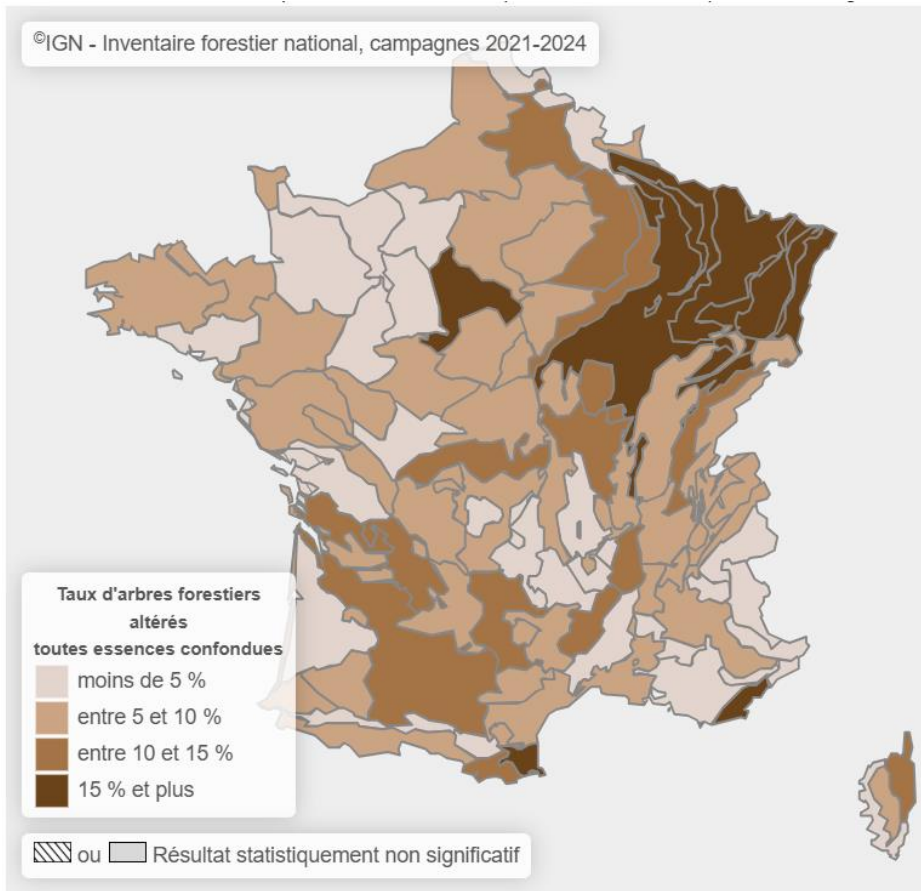
- Value of elements among them (relative approach)
- A quantity related to a reference system (absolute approach)



Source : Chesneau, 2006.

# Properties of visual variables

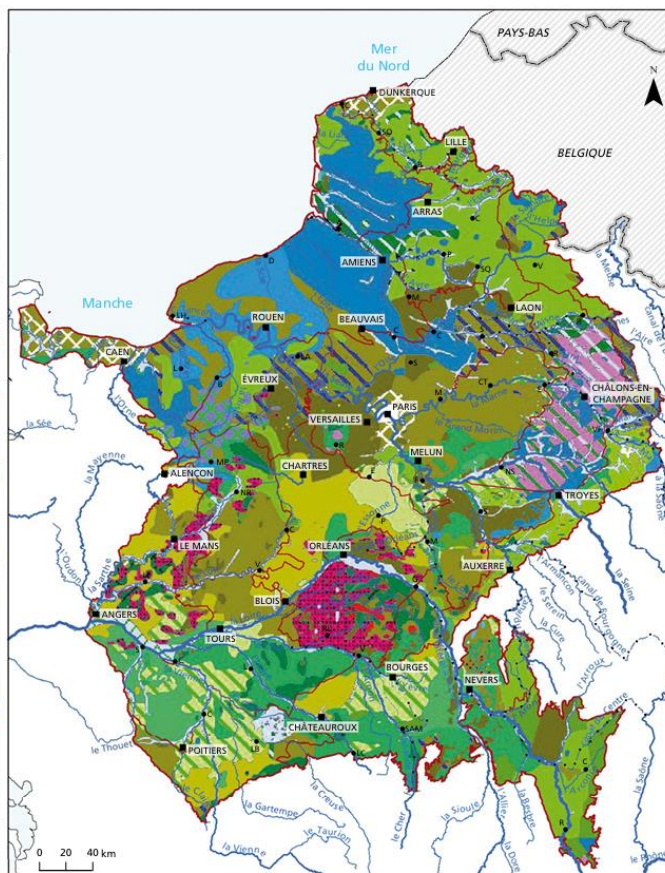
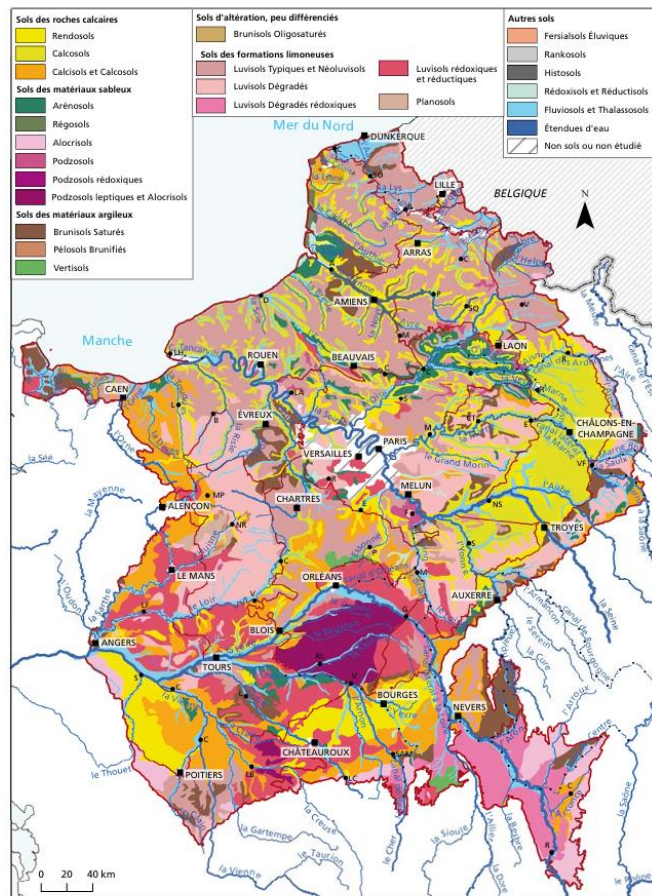
<https://inventaire-forestier.ign.fr/>



Ordre

# Properties of visual variables

<https://inventaire-forestier.ign.fr/>



## Étage planitiaire

- Aulnaie glutineuse avec tourbe
- Aulnaie glutineuse pure
- Aulnaie avec saules / Aulnaie avec peupliers
- Aulnaie avec frêne ou orme

## Étage collinéen

- Chênaie pédonculée et landes acides
- Chênaie pédonculée, mésotrophe à eutrophe
- Chênaie pédonculée, calcicole
- Chênaie pédonculée acide à chêne tauzin
- Chênaie sessile acide
- Chênaie sessile et landes méso-neutrophiles
- Chênaie sessile calcicole
- Chênaie mixte avec chênes sessile et pédonculé à charme et hêtre, landes acides
- Chênaie mixte avec chênes sessile et pédonculé, calcicole
- Chênaie mixte avec chênes sessile et pédonculé mésotrophe à eutrophe, fruticées et pelouses
- Chêne sessile, chêne pédonculé, charme et hêtre
- Bocage de chêne pédonculé, orme, charme et hêtre, avec prairies de fauche et de pâture
- Chênaie pubescente
- Chênaie pubescente mixte à feuillus divers, fruticées et pelouses
- Pin sylvestre
- Pin sylvestre et chênes (pédonculé majoritaire)
- Plantations à pin sylvestre, pin maritime, pin noir et pins méditerranéens
- Pinède silicicole
- Hêtraie-chênaie sessile, acidiphile, friches et pelouses
- Hêtraie-chênaie sessile, neutrophile, friches et pelouses
- Hêtraie hygrosclaphile à tilleul et érable (à sapin dans l'ouest)

Association  
Differentiation

Sources : BD géographique des sols de France au 1/1 000 000 (version 1) © INFOSOL, INRA - simplification © IFN de la carte des sols : BD CARTHAGE© IGN Agences de l'Eau.

Sources : BD ALTI© IGN, BD CARTHAGE© IGN Agences de l'Eau, carte de la végétation de la France : couverture numérique harmonisée au 1/1 000 000 © CNRS-UMR Géographie-cités 8504 (fichier image : fichier vectoriel non accessible).

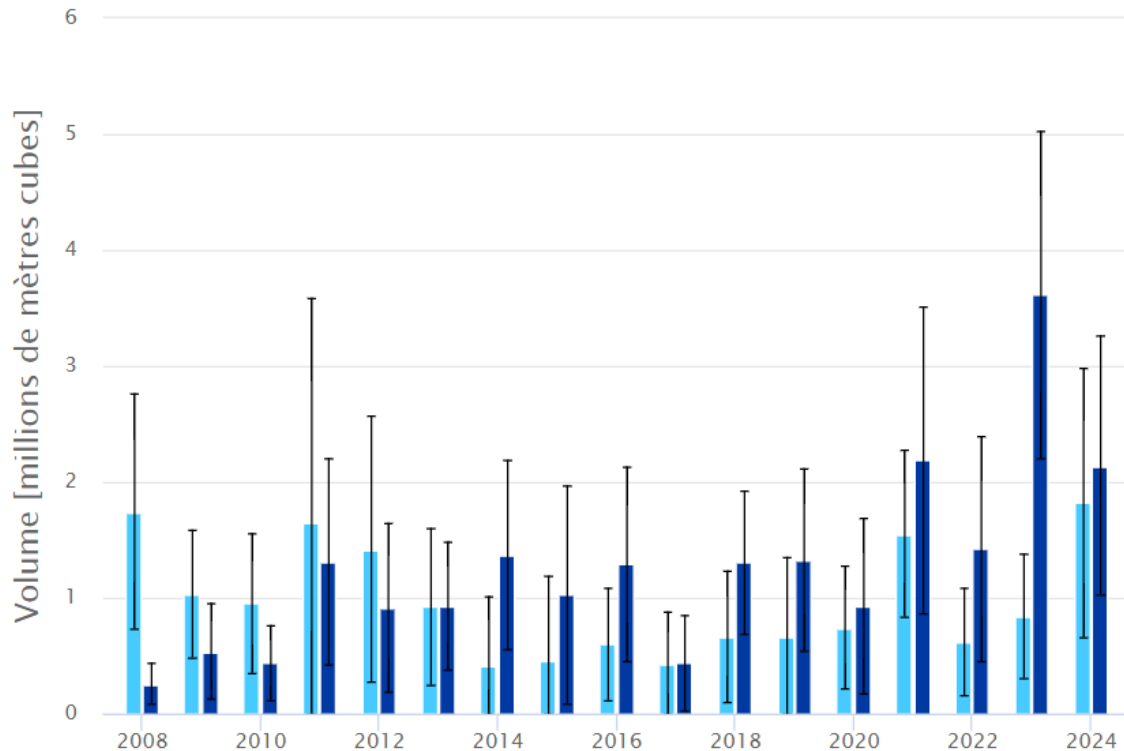


# Properties of visual variables

<https://inventaire-forestier.ign.fr/>

● plus de 5 ans  
● 5 ans ou moins  
— Stock d'arbres morts (± intervalle de confiance)

☰ Exporter la figure



Quantité  
Ordre  
Différenciation

# Graphic semiology rules

Zanin, Trémolo. *Savoir faire une carte*. 2003.

| Type d'implantation | Nature des données                                                                 |                                                                                    |                                                                                                                                                                           |                                                                                                                                                                           |                                                                                   |                                                                                                                                                                           |                                                                                                                                                                               |                                                                                      |                                                                                     |                                                                                      |
|---------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                     | Qualitative                                                                        |                                                                                    |                                                                                                                                                                           |                                                                                                                                                                           |                                                                                   |                                                                                                                                                                           | Quantitative                                                                                                                                                                  |                                                                                      |                                                                                     |                                                                                      |
|                     | Nominale                                                                           |                                                                                    | Ordinale                                                                                                                                                                  |                                                                                                                                                                           |                                                                                   |                                                                                                                                                                           | Relative                                                                                                                                                                      |                                                                                      |                                                                                     | Absolue                                                                              |
| Ponctuelle          | Forme                                                                              | Couleur                                                                            | Taille                                                                                                                                                                    | Valeur                                                                                                                                                                    |                                                                                   | Valeur                                                                                                                                                                    | Couleur                                                                                                                                                                       | Texture                                                                              | Taille                                                                              |                                                                                      |
|                     |   |   | <br>    | <br>    |                                                                                   |                                                                                          |                                                                                             |   |  |                                                                                      |
| Linéaire            | Forme                                                                              | Couleur                                                                            | Taille                                                                                                                                                                    | Valeur                                                                                                                                                                    | Couleur                                                                           | Valeur                                                                                                                                                                    | Couleur                                                                                                                                                                       |                                                                                      | Taille                                                                              |                                                                                      |
|                     |   |   |                                                                                          |                                                                                          |  |                                                                                          |                                                                                            |                                                                                      |  |                                                                                      |
| Zonale              | Couleur                                                                            | Texture                                                                            | Valeur                                                                                                                                                                    | Couleur                                                                                                                                                                   |                                                                                   | Valeur                                                                                                                                                                    | Couleur                                                                                                                                                                       | Taille                                                                               |                                                                                     | Points comptables                                                                    |
|                     |  |  | <br> | <br> |                                                                                   | <br> | <br> |  |                                                                                     |  |

# Graphic semiology rules

Zanin, Trémolo. *Savoir faire une carte*. 2003.

| Type d'implantation | Nature des données                                                                            |                                                                                               |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                             |                                                                                                                                                                                                |                                                                                                                                                                                                 |                                                                                                                                                                                                             |  |
|---------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                     | Qualitative                                                                                   |                                                                                               |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                             | Quantitative                                                                                                                                                                                   |                                                                                                                                                                                                 |                                                                                                                                                                                                             |  |
|                     | Nominale                                                                                      |                                                                                               | Ordinale                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                             | Relative                                                                                                                                                                                       |                                                                                                                                                                                                 | Absolue                                                                                                                                                                                                     |  |
| Ponctuelle          | Forme<br>    | Couleur<br>  | Taille<br><br>Couleur<br><br>Texture<br> | Valeur<br><br>Texture<br> | Valeur<br><br>Couleur<br>   | Texture<br>                                                                                                  | Taille<br>                                                                                                               |  |
| Linéaire            | Forme<br>    | Couleur<br>  | Taille<br><br>Valeur<br>                                                                                                  | Couleur<br>                                                                                                | Valeur<br><br>Couleur<br> |                                                                                                                                                                                                 | Taille<br>                                                                                                               |  |
| Zonale              | Couleur<br> | Texture<br> | Valeur<br><br>Texture<br>                                                                                                 | Couleur<br><br>Grain<br> | Valeur<br><br>Texture<br> | Couleur<br><br>Grain<br> | Taille<br><br>Points comptables<br> |  |

Quantity  
Order  
Differentiation

# Continuity/critic of Bertin graphic semiology

THE VISUAL VARIABLES AND THEIR SYNTACTICS

| ground → figure |                  |             |           |                          |                      |                           |  |
|-----------------|------------------|-------------|-----------|--------------------------|----------------------|---------------------------|--|
|                 |                  | associative | selective | nominal<br>(non ordered) | ordinal<br>(ordered) | numerical<br>quantitative |  |
|                 | location         | Y           | Y         | G                        | G                    | G                         |  |
|                 | size             | N           | Y         | G                        | G                    | G                         |  |
|                 | shape            | Y           | N         | G                        | P                    | P                         |  |
|                 | orientation      | Y           | Y         | G                        | M                    | M                         |  |
|                 | color hue        | Y           | Y         | G                        | M                    | M                         |  |
|                 | color value      | N           | Y         | P                        | G                    | M                         |  |
|                 | texture          | Y           | Y         | G                        | M                    | M                         |  |
|                 | color saturation |             |           | P                        | G                    | M                         |  |
|                 | arrangement      |             |           | M                        | P                    | P                         |  |
|                 | crispness        |             |           | P                        | G                    | P                         |  |
|                 | resolution       |             |           | P                        | G                    | P                         |  |
|                 | transparency     |             |           | M                        | G                    | P                         |  |

visual variable variations Y=yes; N=no; G=good; M=marginal; P=poor; hatched=n/a

## Extension of graphic semiology

*Bertin 1987, MacEachren 1995, Brewer 2005, MacEachren et al. 2012, Roth 2015, etc.*

## Evaluation of visual variables proposed by graphic semiology

*Cleveland 1985, Cleveland & McGill 1988, Mackinlay 1986, Wilkinson 1999*

# Continuity/critic of Bertin graphic semiology

## THE VISUAL VARIABLES AND THEIR SYNTACTICS

| ground | figure |                  | associative | selective | nominal<br>(non ordered) | ordinal<br>(ordered) | numerical<br>(quantitative) |
|--------|--------|------------------|-------------|-----------|--------------------------|----------------------|-----------------------------|
|        |        | location         | Y           | Y         | G                        | G                    | G                           |
|        |        | size             | N           | Y         | G                        | G                    | G                           |
|        |        | shape            | Y           | N         | G                        | P                    | P                           |
|        |        | orientation      | Y           | Y         | G                        | M                    | M                           |
|        |        | color hue        | Y           | Y         | G                        | M                    | M                           |
|        |        | color value      | N           | Y         | P                        | G                    | M                           |
|        |        | texture          | Y           | Y         | G                        | M                    | M                           |
|        |        | color saturation |             |           | P                        | G                    | M                           |
|        |        | arrangement      |             |           | M                        | P                    | P                           |
|        |        | crispness        |             |           | P                        | G                    | P                           |
|        |        | resolution       |             |           | P                        | G                    | P                           |
|        |        | transparency     |             |           | M                        | G                    | P                           |

visual variable variations Y=yes; N=no; G=good; M=marginal; P=poor; hatched=n/a

## Extension of graphic semiology

Bertin 1987, MacEachren 1995, Brewer 2005, MacEachren et al. 2012, Roth 2015, etc.

## Evaluation of visual variables proposed by graphic semiology

Cleveland 1985, Cleveland & McGill 1988, Mackinlay 1986, Wilkinson 1999

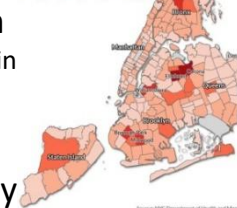
## Arrival of computer graphics, Web cartography, crowdsourced cartography

- animation, interaction, sound, 3D
- New visual variables using animation (Appearance, Duration, Speed, Movement, Blinkin Arnaud et al. 2009
- New creators of maps, from other domains than cartography

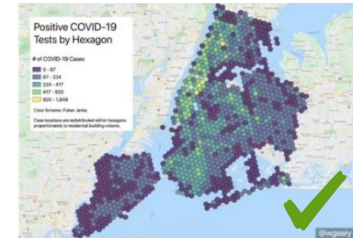
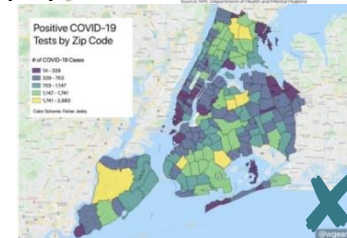
→ new uses of visual variables

Mericskay, 2017

Coronavirus cases in New York City  
A ZIP Code by ZIP Code map of which neighborhood have the most cases of COVID-19 as of April 1, 2020



Coronavirus Cases in New York City

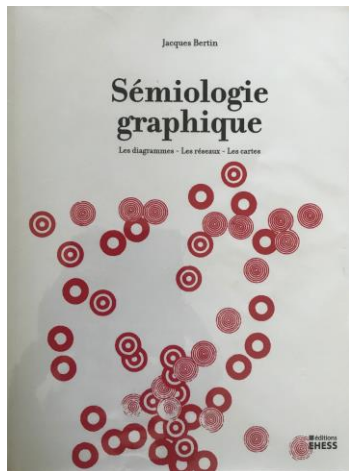


## New approaches from computer science, dataviz

New nomenclatures to classify visual variables

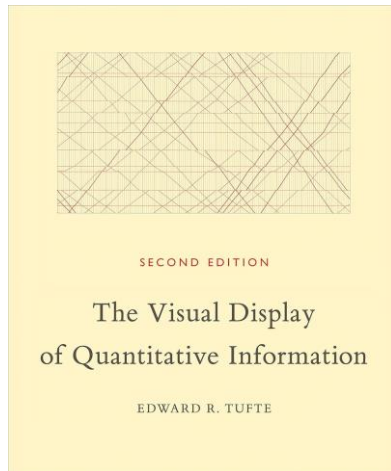
Munzner 2014

# Symbolic language



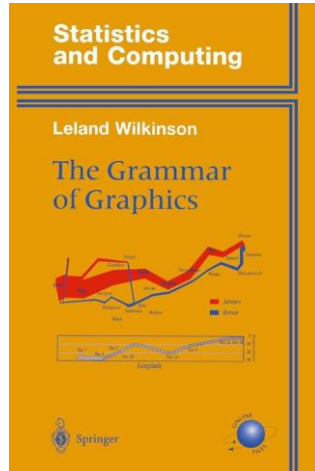
1967

Jacques Bertin



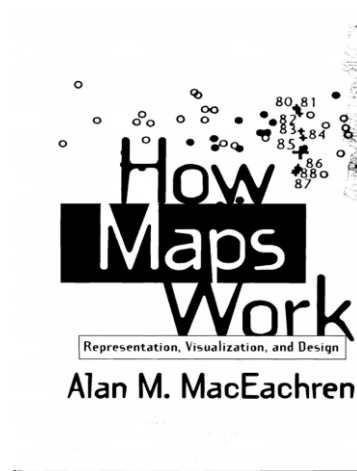
1983

Edward Tufte



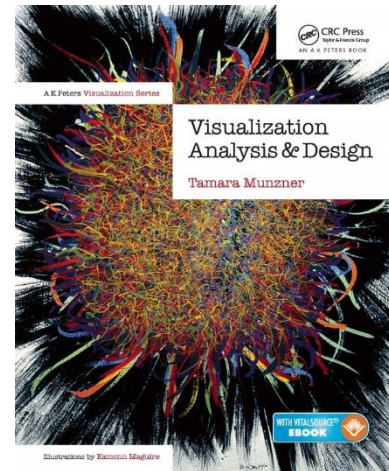
2001

Leland Wilkinson



2004

Alan MacEachren



2014

Tamara Munzner



# Munzner visualization taxonomy

## Mark

Basic graphical element in an image.

Classified according to the number of spatial dimensions:

### 0 dimensions

→ Points



### 1 dimension

→ Lines



### 2 dimensions

→ Areas



## Channel

Means to control the visual appearance of a mark.

Some examples of channels:

→ Position

→ Horizontal



→ Vertical



→ Both



→ Color



(hue, saturation, luminance)

→ Shape



→ Tilt



→ Size

→ Length



→ Area



→ Volume



# Visual Channels

Identity: **what** is something or **where** it is

**Differentiation & Association**

→ Shape



→ Color (hue)



Motion pattern



Spatial region



Magnitude: **how much** is something

**Order & Quantity**

→ Position

→ Horizontal



→ Vertical



→ Both



Color (luminance, saturation)



→ Tilt



→ Size

→ Length



→ Area



→ Volume



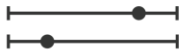
*Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014.*



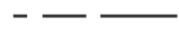
# Visual Channels

Channels: Expressiveness Types and Effectiveness Ranks

## ➔ Magnitude Channels: Ordered Attributes

Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

Color luminance 

Color saturation 

Curvature 

Volume (3D size) 

## ➔ Identity Channels: Categorical Attributes

Spatial region 

Color hue 

Motion 

Shape 

Most

Effectiveness

Least

Same

Same

# Visual Channels

Channels: Expressiveness Types and Effectiveness Ranks

➔ **Magnitude Channels: Ordered Attributes**

Position on common scale



Position on unaligned scale



Length (1D size)



Area (2D size)



Color luminance



Color saturation



Same

Most

Effectiveness

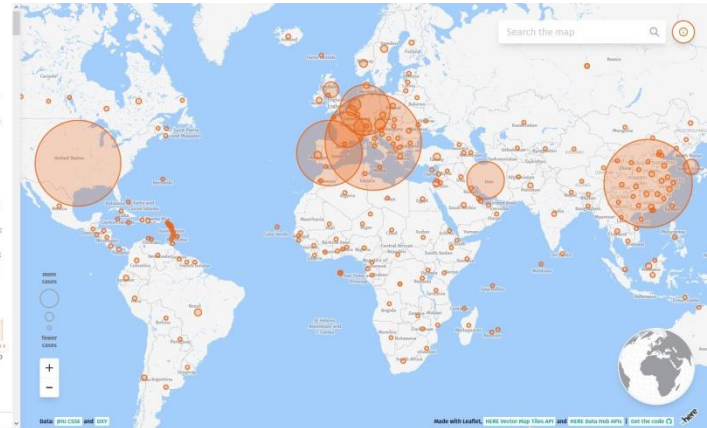
Least



## Tracking Coronavirus COVID-19

PLEASE NOTE:  
Due to changing datasets provided by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University, data on recoveries is currently unavailable. Once the recovery data becomes available, we will update this map accordingly.

The first case of the new Coronavirus COVID-19 was recorded on 31 December in Wuhan, China (Hubei). Since then, 46,593 confirmed cases have been recorded worldwide. This visualization shows the near real-time status based on data from the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University and WHO. Data currently available on the following zoom levels: City level - US, Canada and Australia; Province level - China; Country level - other countries. To read more about this map, see [How we built an interactive map displaying the COVID-19 outbreak](#).



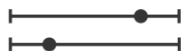
In a map,  
it is easier to **compare quantities**  
with symbols' **area**  
than with symbols' **color**

# Visual Channels

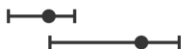
Channels: Expressiveness Types and Effectiveness Ranks

## ➔ Magnitude Channels: Ordered Attributes

Position on common scale



Position on unaligned scale



Length (1D size)



Area (2D size)



Color luminance



Color saturation



Same

▲ Most

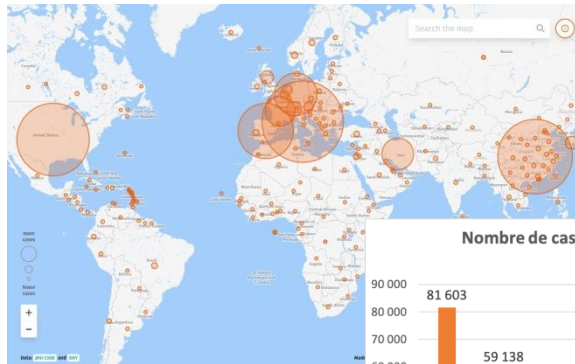
Effectiveness

▼ Least

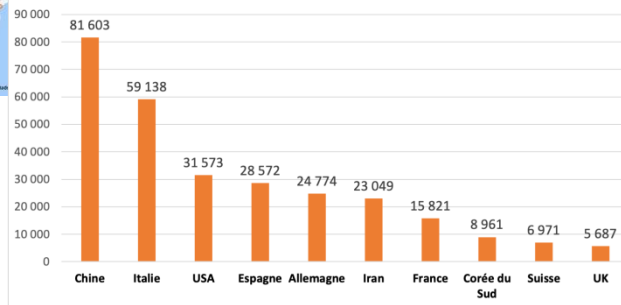
It's easier to **compare quantities**

Through the **vertical position of aligned symbols** in a **bar chart**

Than through symbol's **area** into a **map**



Nombre de cas de Covid-19 depuis le début de l'épidémie  
Le top 10



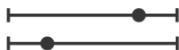
Tamara Munzner. *Visualization Analysis and Design. A K Peters Visualization Series*, CRC Press, 2014.

# Visual Channels

Channels: Expressiveness Types and Effectiveness Ranks

## ➔ Magnitude Channels: Ordered Attributes

Position on common scale



Position on unaligned scale



Length (1D size)



Area (2D size)



Color luminance



Color saturation



Same

Most

Effectiveness

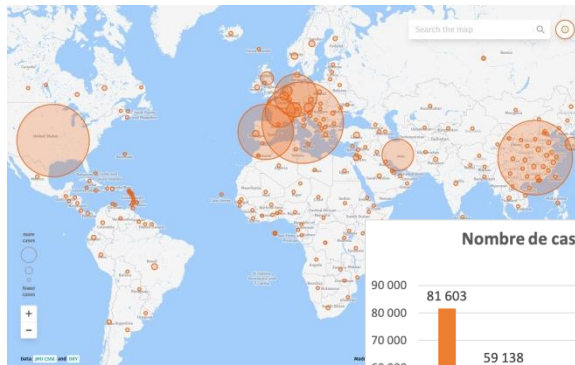
Least

It's easier to **compare quantities**

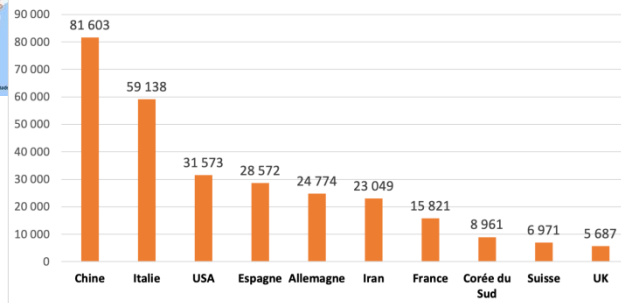
Through the **vertical position of aligned symbols** in a **bar chart**

Than through symbol's **area** into a **map**

The spatial distribution is not always the most pertinent information to represent



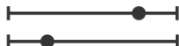
Nombre de cas de Covid-19 depuis le début de l'épidémie  
Le top 10



# Visual Channels

Channels: Expressiveness Types and Effectiveness Ranks

## ➔ Magnitude Channels: Ordered Attributes

Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

Color luminance 

Color saturation 

Curvature 

Volume (3D size) 

## ➔ Identity Channels: Categorical Attributes

Spatial region 

Color hue 

Motion 

Shape 

Most

Effectiveness

Least

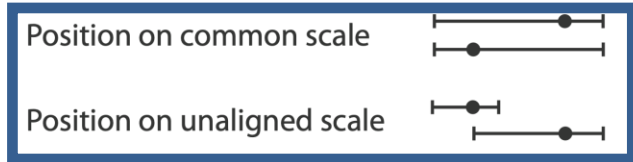
Same

Same

# Visual Channels

Channels: Expressiveness Types and Effectiveness Ranks

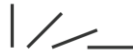
## ➔ Magnitude Channels: Ordered Attributes



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)



## ➔ Identity Channels: Categorical Attributes



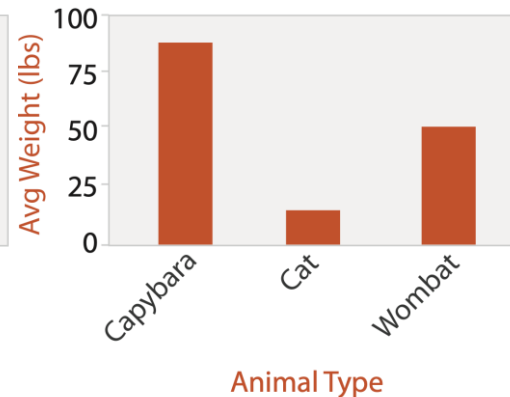
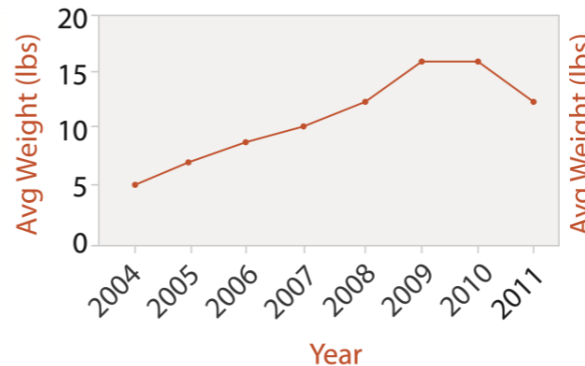
Color hue



Motion



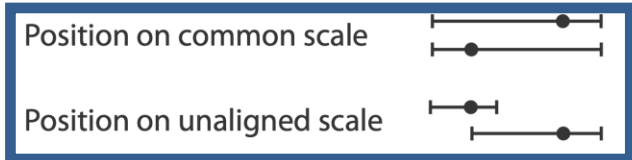
Shape



# Visual Channels

Channels: Expressiveness Types and Effectiveness Ranks

## ➔ Magnitude Channels: Ordered Attributes



Length (1D size)

Tilt/angle

Area (2D size)

Depth (3D position)

Color luminance

Color saturation

Curvature

Volume (3D size)

Most

Effectiveness

Least

## ➔ Identity Channels: Categorical Attributes

Spatial region

Color hue

Motion

Shape

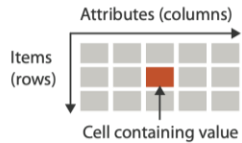
Symbol's graphic position is the **most efficient** channel among magnitude and identity channels, and the **only one to cover both types**

**Only true for 2D visualization.**

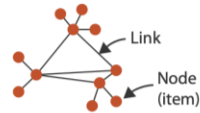
# Different data

## → Dataset Types

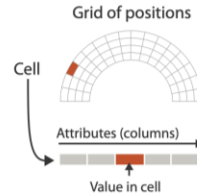
### → Tables



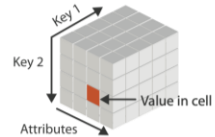
### → Networks



### → Fields (Continuous)



### → Multidimensional Table



### → Trees



### → Geometry (Spatial)

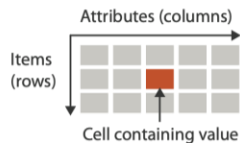




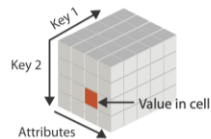
# Different data → Different approaches

## → Dataset Types

### → Tables



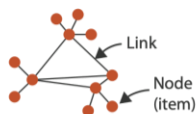
### → Multidimensional Table



### → Geometry (Spatial)



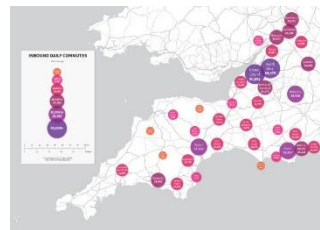
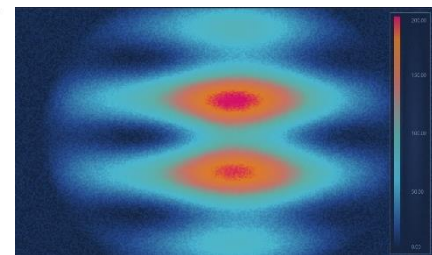
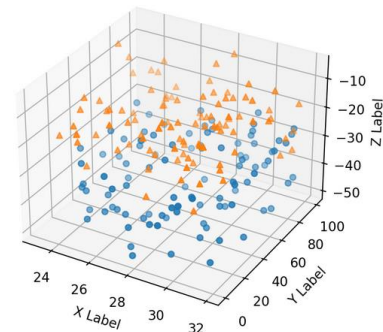
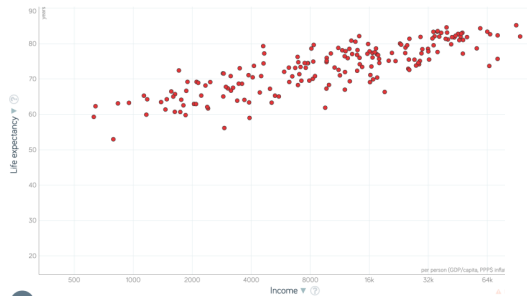
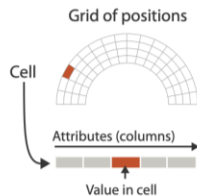
### → Networks



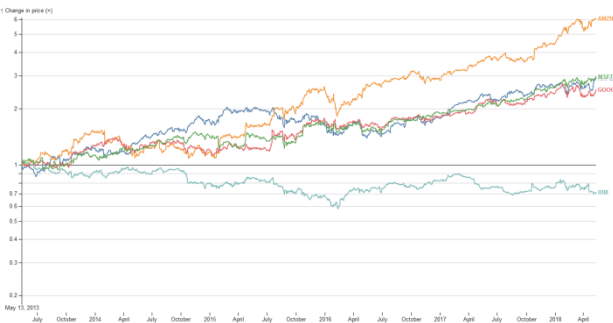
### → Trees



### → Fields (Continuous)

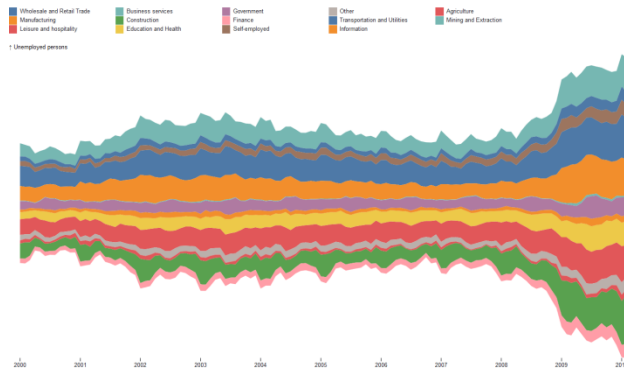


# ≠ approaches for ≠ needs



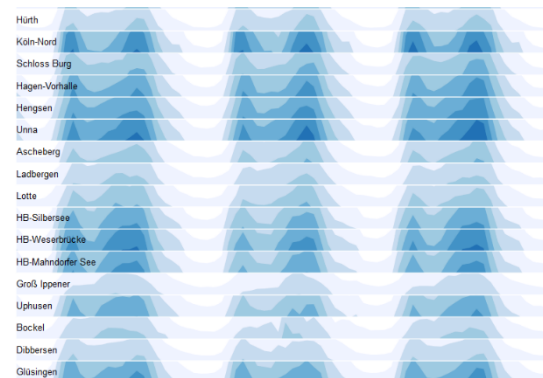
Line Chart

<https://observablehq.com/@d3/index-chart>



Stacked Area Chart

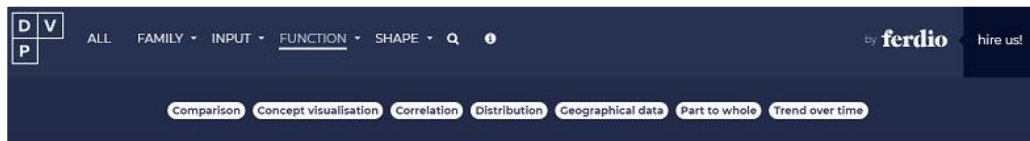
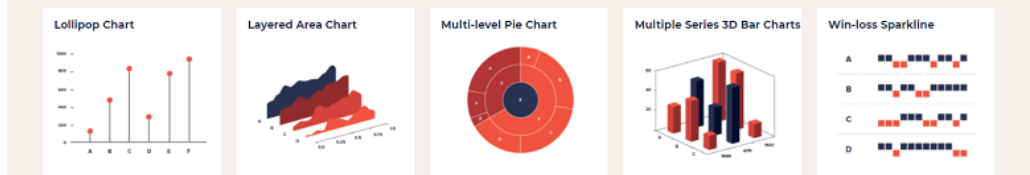
<https://observablehq.com/@d3/streamgraph>



Horizon Chart

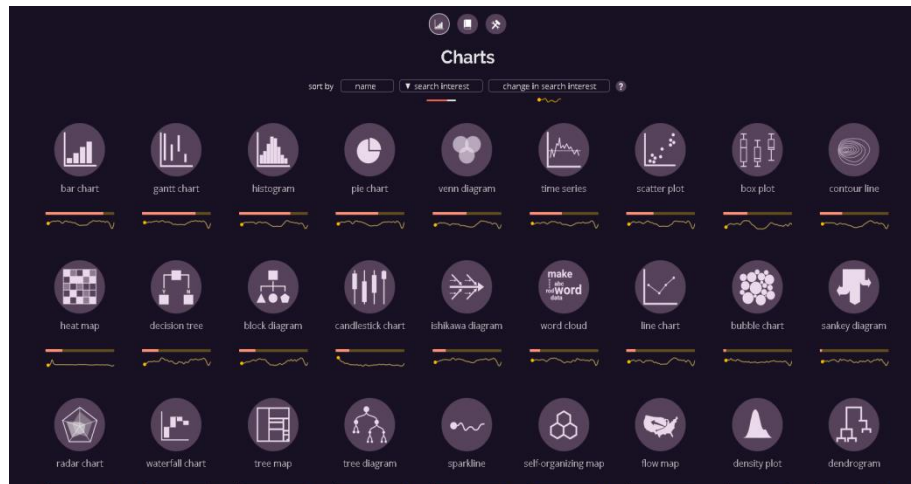
<https://observablehq.com/@d3/horizon-chart>

# ≠ approaches for ≠ needs

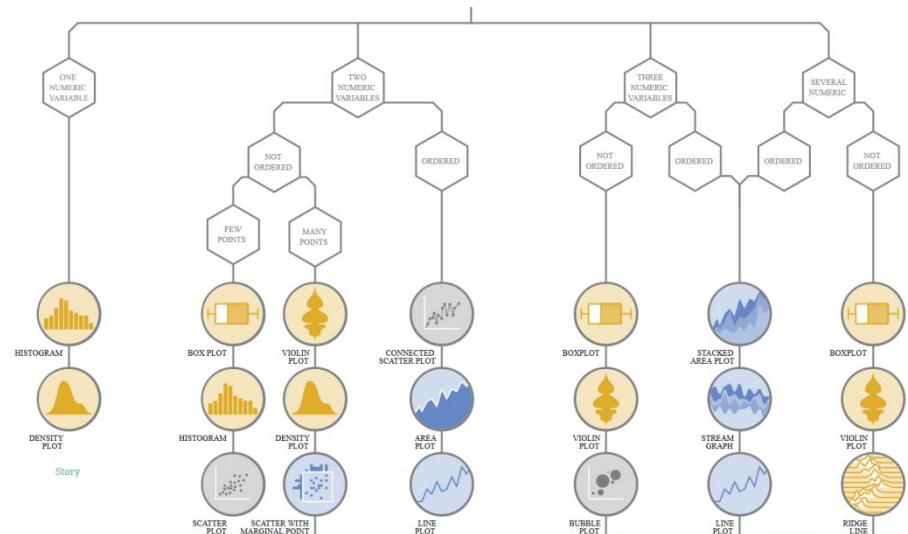


Data Viz Project: <https://datavizproject.com/>

# ≠ approaches for ≠ needs



<http://visualizationuniverse.com/charts/>



<https://www.data-to-viz.com/>

# ≠ approaches for ≠ needs

<https://www.data-to-viz.com/>

What kind of data do you have? Pick the main type using the buttons below. Then let the decision tree guide you toward your graphic possibilities.

Numeric

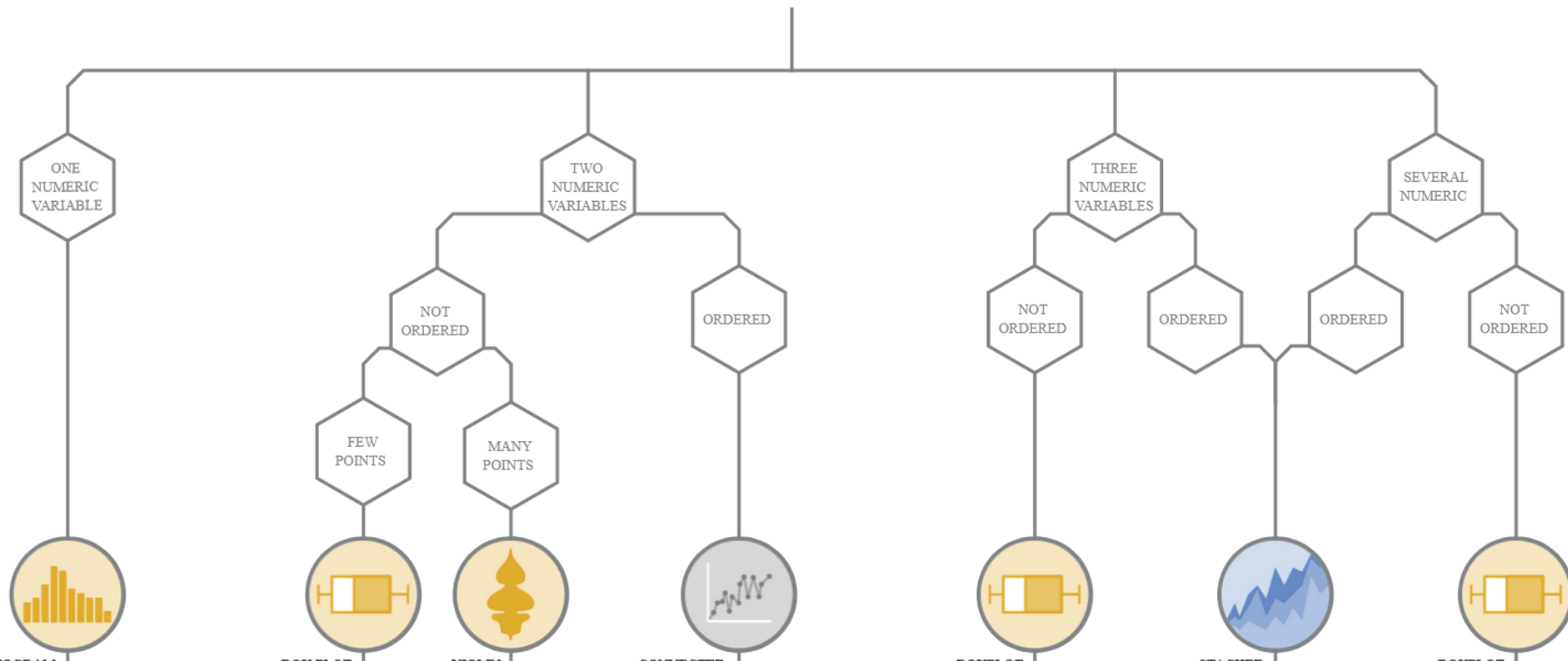
Categoric

Num & Cat

Maps

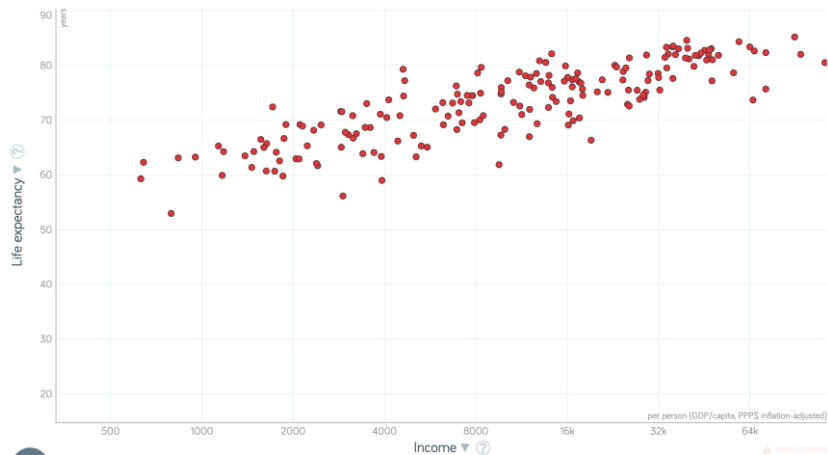
Network

Time series



# Visualizing tables

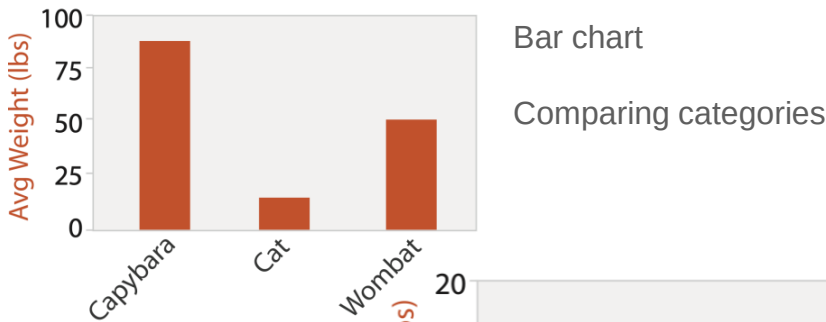
## Visualizing raw data distribution



Scatterplot

Identify correlation between two attributes  
Identify clusters

## Visualizing data distribution agregated according to key- attribute modalities

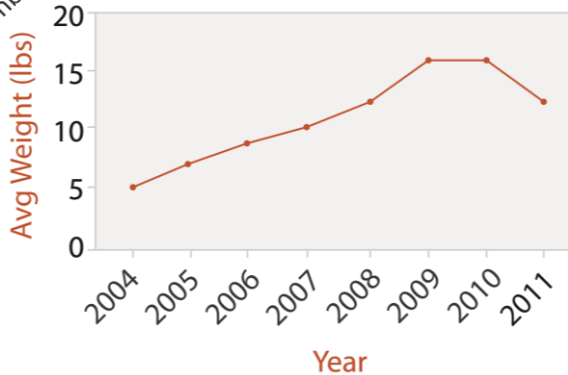


Bar chart

Comparing categories

Line chart

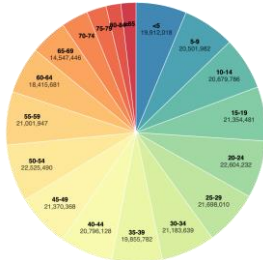
Identify tendencies in  
evolution according  
one ordered attribute



Year

Pie chart

Evaluate proportion  
Part-to-whole  
relationships



# Visualizing tables

Be careful with the introduction of biases through scales



*Truncating the Y-Axis: Threat or Menace? Correll, Bertini, & Franconeri, CHI 2020.*

## Be careful with the introduction of biases through scales

→ Ordering Direction

- Sequential  
→
- Diverging  
← →
- Cyclic  
↻

[illegible]

Vismon: Facilitating Risk Assessment and Decision Making In Fisheries Management. Booshehrian, Möller, Peterman, and Munzner. Technical Report TR 2011-04, Simon Fraser University, School of Computing Science, 2011.

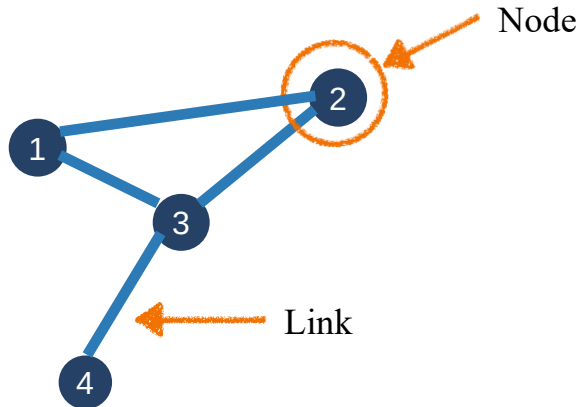


# Visualizing networks

## Wide meaning

- Physical network
- Origin-destination flows
- Social network
- Filiation between entities
- Evolution between different states of an entity

## Topological representation

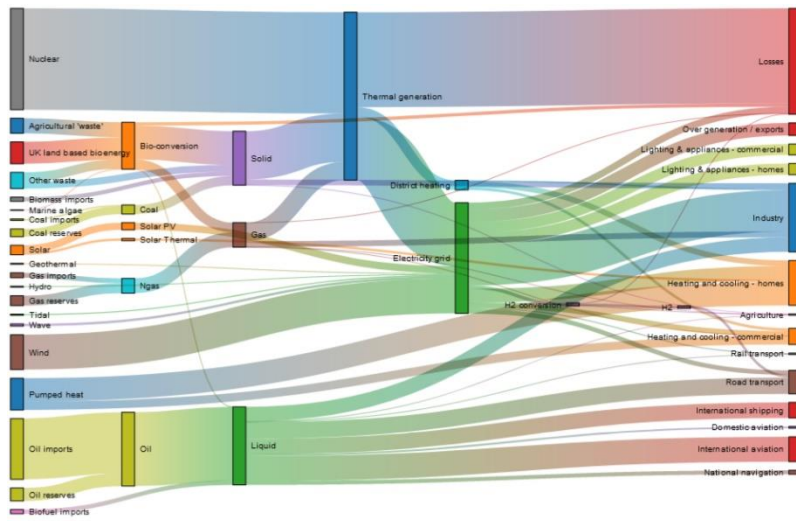
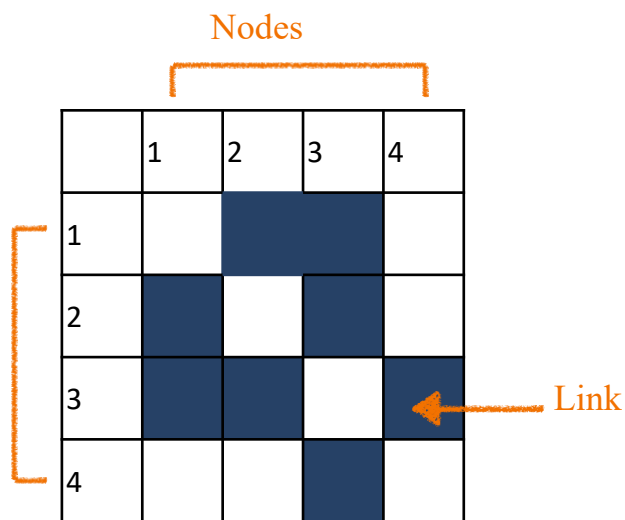


Exploring network structure;  
identifying paths

## Adjacency matrix Origin destination matrix

Deeper analysis of  
relationships between  
individuals

Nodes



Sankey Chart  
Representing filiation,  
evolution, exchanges  
between individuals

<https://observablehq.com/@d3/sankey/2?intent=fork>

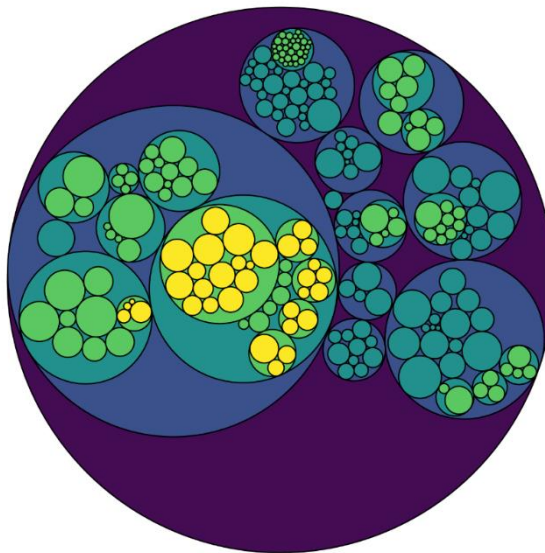
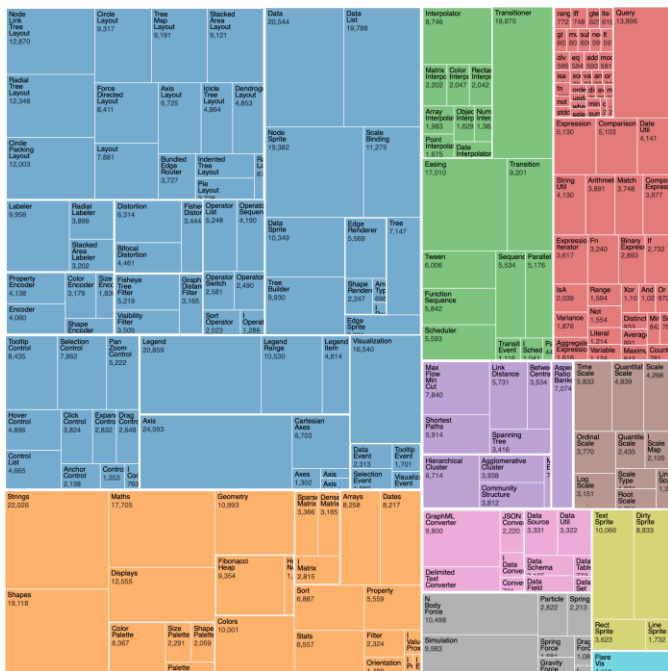


# Visualizing hierarchical data

## Wide meaning

- Hierarchical network
- Data categorized at multiple levels

# Containment



## Circular Packing

<https://r-graph-gallery.com/314-custom-circle-packing-with-several-levels.html>

More adapted to represent proportions of the different categories at each level

## Treemap

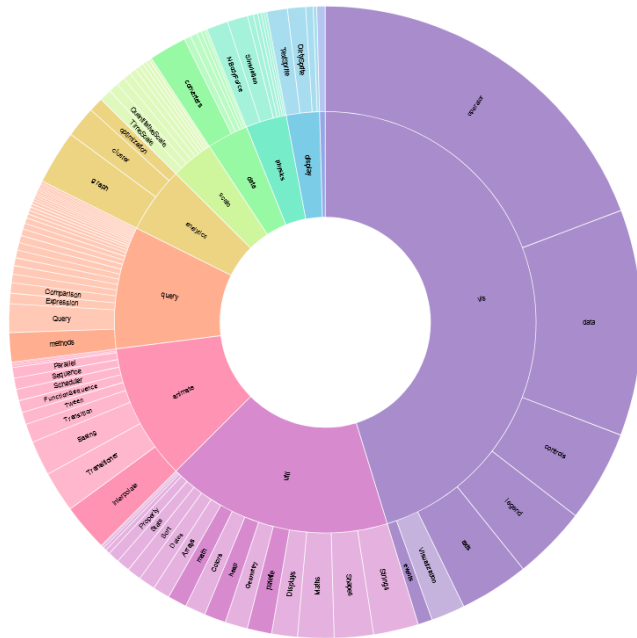
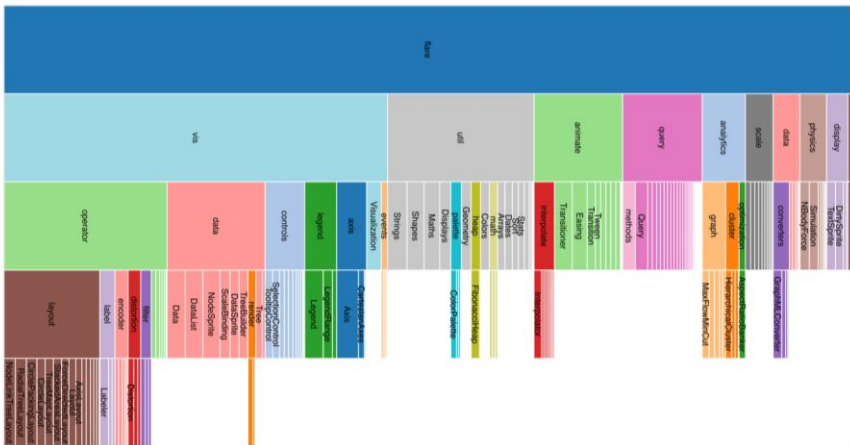
<https://observablehq.com/@d3/treemap>

# Visualizing hierarchical data

## Wide meaning

- Hierarchical network
- Data categorized at multiple levels

## Position



Sunburst  
<https://observablehq.com/@d3/zoomable-sunburst>

Combine the two approaches

Icicle Plot  
Munzner 2014

Technical solutions

# Technical solutions

## Developpers solutions



## Non-developoppers solution



# Technical solutions

## Developpers solutions

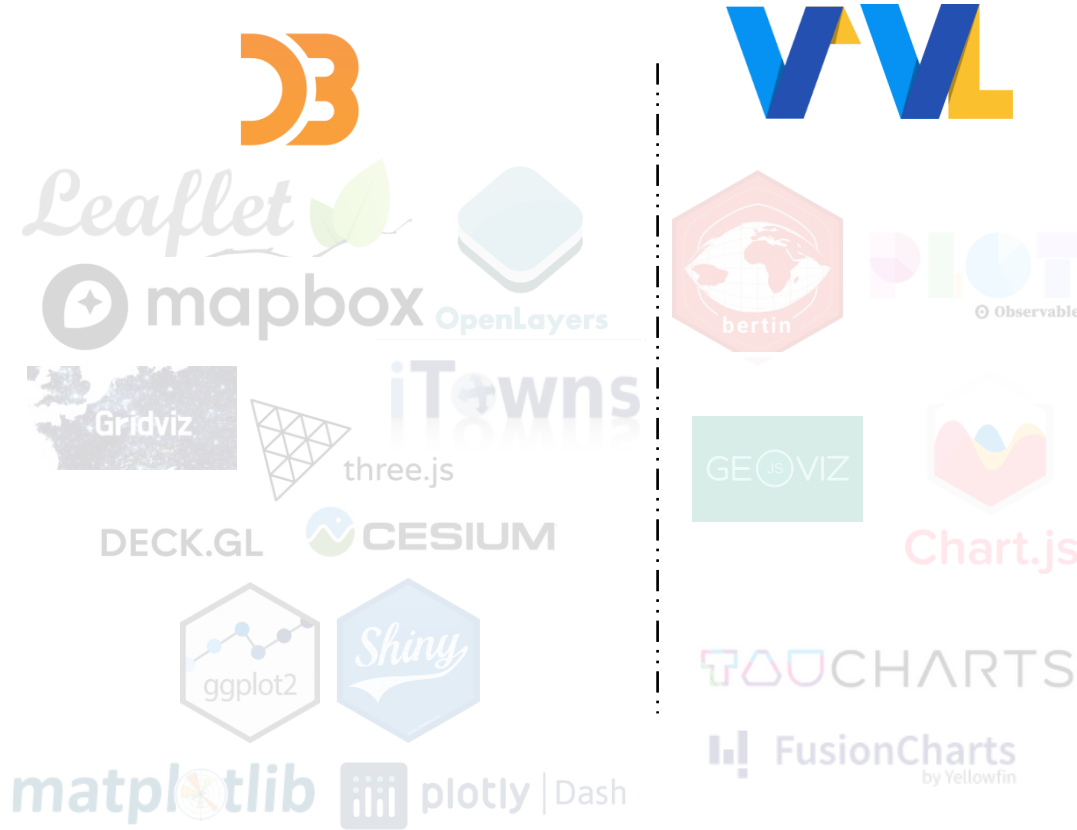


## Non-developpers solution



# Technical solutions

## Developpers solutions



## Non-developpers solution

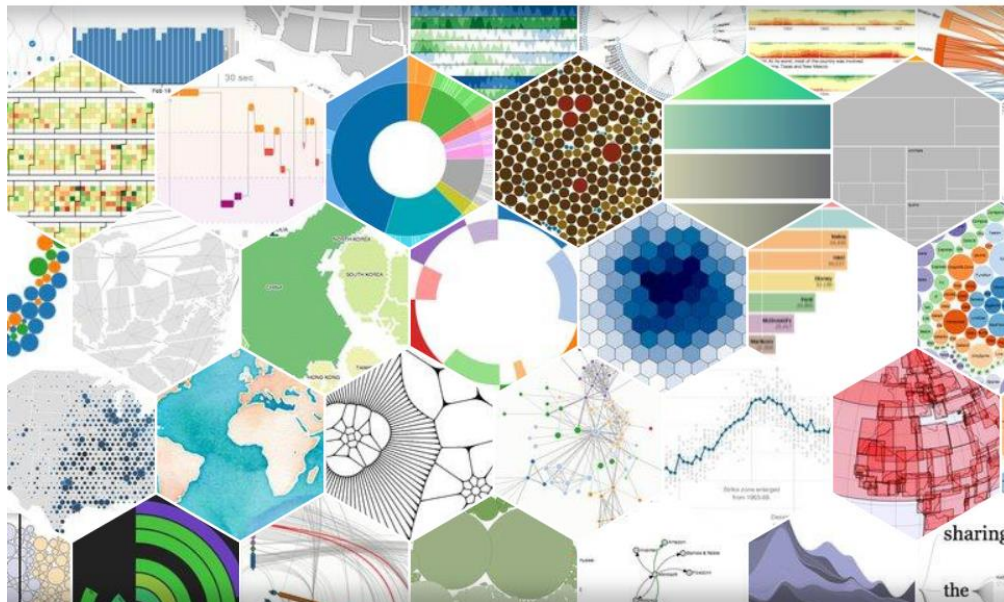




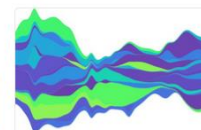
# Technical solutions

D3 <https://d3js.org/>

 Data-Driven Documents



Stacked-to-grouped bars



Streamgraph transitions



Smooth zooming



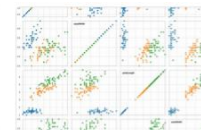
Zoom to bounding box



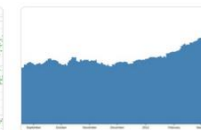
Sequences sunburst



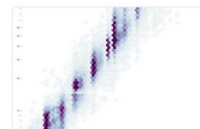
Brushable scatterplot



Brushable scatterplot matrix



Pannable chart



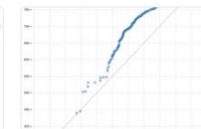
Hexbin



Hexbin (area)



Hexbin map



Q-Q plot



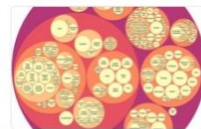
Treemap



Cascaded treemap



Nested treemap



Circle packing

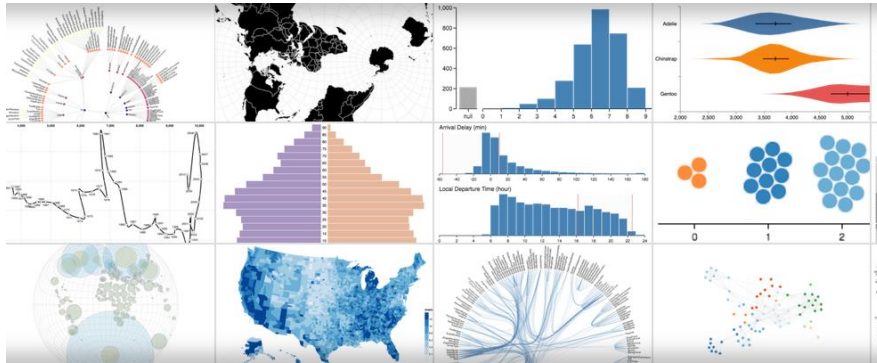
Reference Javascript library for 2D visualization  
Allows vectorial drawing in HTML document

Mike Bostock  
Former NY Times visualization teams

# Technical solutions

## Vega

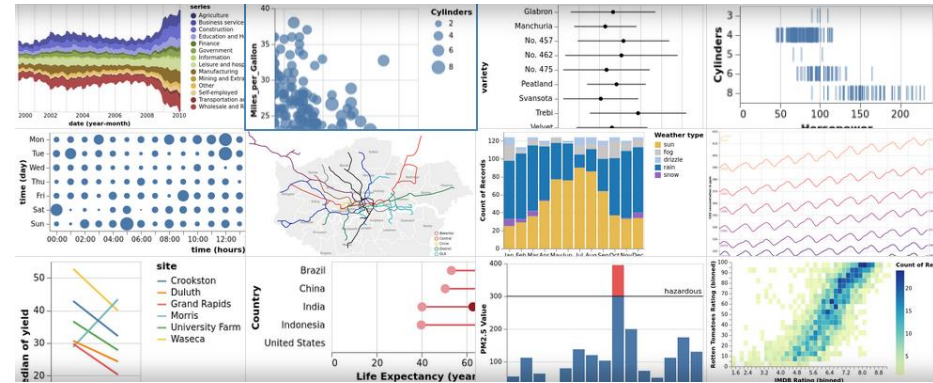
<https://vega.github.io/vega/>



Based on D3, following ‘Visualization grammar’ principles  
Declarative language for implementing different visualization approaches

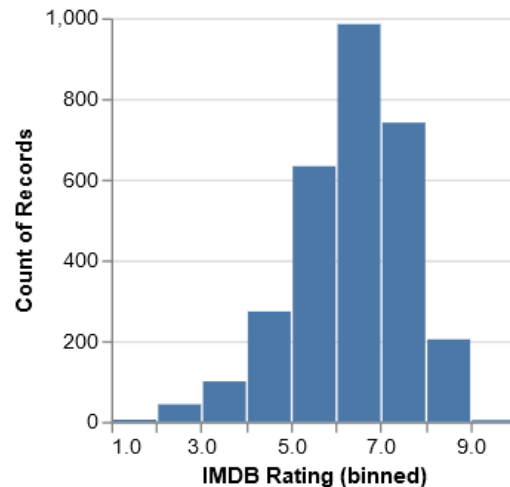
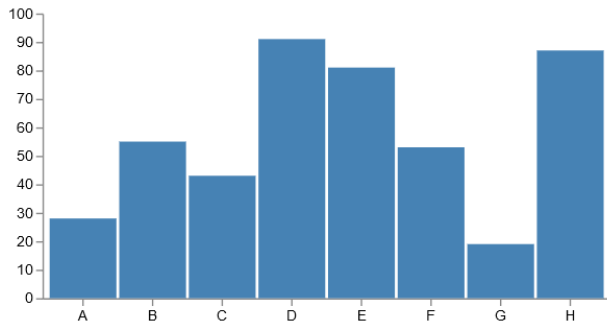
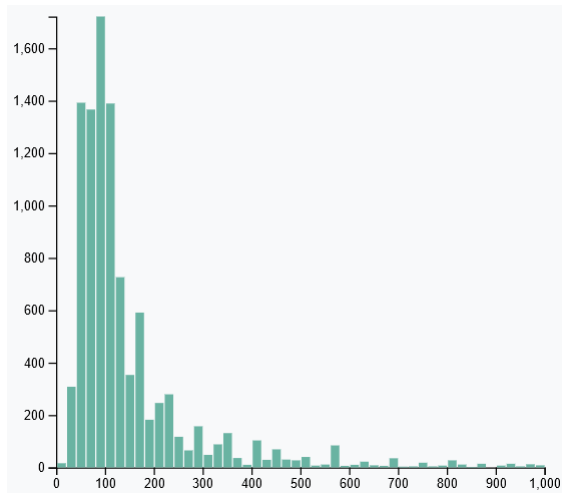
## Vega-Lite

<https://vega.github.io/vega-lite/>

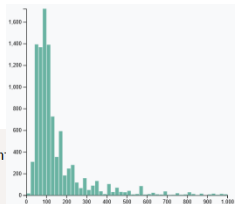


Lighter version  
Easier to use.

# Technical solutions



# Technical solutions



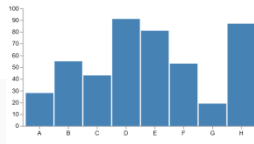
```
svg.append("g")
  .attr("transform", "translate(0," + height)
  .call(d3.axisBottom(x));

// set the parameters for the histogram
var histogram = d3.histogram()
  .value(function(d) { return d.price; }) // I need to give the vector of
  .domain(x.domain()) // then the domain of the graphic
  .thresholds(x.ticks(70)); // then the numbers of bins

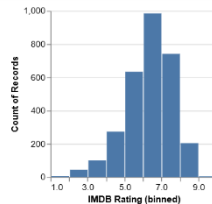
// And apply this function to data to get the bins
var bins = histogram(data);

// Y axis: scale and draw:
var y = d3.scaleLinear()
  .range([height, 0]);
y.domain([0, d3.max(bins, function(d) { return d.length; })]); // d3.his
svg.append("g")
  .call(d3.axisLeft(y));

// append the bar rectangles to the svg element
svg.selectAll("rect")
  .data(bins)
  .enter()
  .append("rect")
  .attr("x", 1)
  .attr("transform", function(d) { return "translate(" + x(d.x0) + "," + y
  .attr("width", function(d) { return x(d.x1) - x(d.x0) - 1; })
  .attr("height", function(d) { return height - y(d.length); })
  .style("fill", "#69b3a2")
```



```
"marks": [
  {
    "type": "rect",
    "from": {"data": "table"},
    "encode": {
      "enter": {
        "x": {"scale": "xscale", "field": "category"},
        "width": {"scale": "xscale", "band": 1},
        "y": {"scale": "yscale", "field": "amount"},
        "y2": {"scale": "yscale", "value": 0}
      },
      "update": {
        "fill": {"value": "steelblue"}
      },
      "hover": {
        "fill": {"value": "red"}
      }
    }
  }
],
```

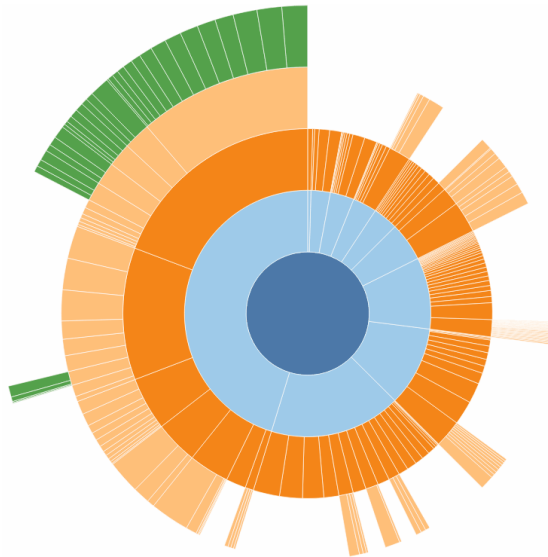


```
{
  "$schema": "https://vega.github.io/
  "data": {"url": "data/movies.json"},
  "mark": "bar",
  "encoding": {
    "x": {
      "bin": true,
      "field": "IMDB Rating"
    },
    "y": {"aggregate": "count"}
  }
}
```

Complexity



# Technical solutions



## Possibilities

## Complexity

# Practical exercise with Vega-Lite

Vega-Lite

Examples

Tutorials

Documentation

Usage

Ecosystem

GitHub

Try Online

## Vega-Lite – A Grammar of Interactive Graphics

